Green and Digital Technology: Exposure, Substitution and Complementarity*

Sidharth Rony† UNU-MERIT
Tommaso Ciarli UNU-MERIT and SPRU
Önder Nomaler UNU-MERIT

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Abstract

Most exposure measures record how close a technology is to an occupation's work and leave open whether it takes the work over or supports it. We measure that direction for the digital and green technologies diffusing in Europe and test whether it is reflected in employment. A large language model, prompted as a chief executive, a manager or a worker, rates substitution and complementarity separately for some 12,000 combinations of a technology, an occupation and an industry observed in European employment data. A companion paper shows that a model run with the same prompt design and sampling settings matches the rating distributions of domain experts, though not rating by rating and least well on substitution. The digital technologies come from a new taxonomy of 184 technologies built from Wikipedia and patents, and the green ones from the OECD ENV-TECH classification. Exposure is measured with the same text, embeddings and aggregation in both domains and correlates with the published indices that measure the same technology, exposure to artificial intelligence with the five AI and generative-AI indices and to robotics with the robot index. Substitution is rated above complementarity in only about one combination in ten, and direction is set by the technology and the occupation. The industry changes its strength but rarely its sign. In the EU Labour Force Survey, occupations with more digital complementarity content grew faster over 2011–2023 than comparable occupations, and those with more digital substitution content grew more slowly.

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†Email: rony@merit.unu.edu

1 Introduction

When automated teller machines took over the core task of bank tellers, the number of tellers rose, as cheaper branch operation led banks to open more branches (Bessen, 2016). Knowing that a technology touches the work of an occupation does not tell us which way it acts. It may take the work over, or support the people who keep doing it. In the task framework the employment effect of a technology nets a displacement effect against productivity and reinstatement effects (Acemoglu and Restrepo, 2019), and the distinction between automation and augmentation organises the measurement of new work (Autor, Chin, Salomons, and Seegmiller, 2024). The exposure measures used to study technology and work at scale record proximity and are silent on the direction in which a technology acts.

Existing measures also concentrate on artificial intelligence (AI), robots and software (Felten, Raj, and Seamans, 2018; Webb, 2020; Acemoglu and Restrepo, 2020). The green transition changes the task content of work no less directly, and the green-skills literature measures how green the tasks of an occupation are (Consoli, Marin, Marzucchi, and Vona, 2016; Vona, Marin, and Consoli, 2019), but no patent-based, technology-specific exposure measure exists for green technologies. Most measures return one score per occupation for a broad domain. Webb (2020) distinguishes three technologies and Prytkova, Petit, Li, Chaturvedi, and Ciarli (2024) a set of emerging digital technologies, but we are not aware of a measure built on an explicit hierarchy of technologies that separates, for instance, computer vision from language processing within the same domain.

Direction is a judgement about how a technology affects a particular kind of work. Expert-based measures score automatability alone, through a rubric applied to tasks by crowd workers (Brynjolfsson, Mitchell, and Rock, 2018) or through experts classifying occupations (Frey and Osborne, 2017). Recent work infers direction. Gmyrek, Berg, and Bescond (2023) classify occupations by the mean and spread of a language model's exposure scores across their tasks, Autor et al. (2024) separate augmentation from automation innovations in patent text, and Kogan, Papanikolaou, Schmidt, and Seegmiller (2023) construct labour-saving and labour-augmenting exposure from patent and task text. Asking experts for the direction directly would be prohibitively costly at the scale needed to cover technologies, occupations and industries jointly. Large language models make such a survey feasible. Prompted with a respondent's profile, they reproduce the opinion distributions of some human samples (Argyle, Busby, Fulda, Gubler, Rytting, and Wingate, 2023), though not of every group, and their answers depend on prompt wording and change over time (Santurkar, Durmus, Ladhak, Lee, Liang, and Hashimoto, 2023; Bisbee, Clinton, Dorff, Kenkel, and Larson, 2024). In our companion paper (Rony and Ciarli, 2026), a model prompted as a respondent re-answers a completed Delphi survey in which some 600 domain experts rated whether specific technologies would substitute for, complement, routinise or make more efficient specific work tasks. For each construct, its rating distribution is close to that of the experts. It does not reproduce individual ratings, and it is weakest on substitution, which it rates about a quarter of a scale point lower than the experts do. A model with open weights gives an assessment that others can rerun. Whether this assessment is more accurate than that of the experts cannot be established without the outcome they were

forecasting. We also use a large language model (LLM) as a judge (Gilardi, Alizadeh, and Kubli, 2023; Gu, Jiang, Shi et al., 2024) in constructing the technology units and the exposure measure. It decides whether a candidate Wikipedia page describes a digital technology, and it later reviews the pages of every technology and the links between patents and occupations or industries on which exposure rests.

This paper measures the direction of technological change for the digital and the green technologies diffusing in Europe and tests whether it is reflected in employment. We select some 12,000 combinations of a technology, a three-digit occupation and an industry section in which exposure is high and the occupation-industry pair is observed in European employment data. For each combination, a language model rates two constructs separately. Substitution is the technology taking over the work so that fewer workers are needed, and complementarity the technology supporting the workers who keep doing it. It rates each construct fifteen times, under the three workplace roles of a chief executive, a manager and a worker. Each agreeing rating is mapped to the fraction of the exposed task content that its scale label names, each disagreeing rating and the zero to none, and the fraction is multiplied by exposure, which gives every occupation and industry two separate measures, its substitution content and its complementarity content. The digital technologies come from a taxonomy that we build from Wikipedia and patents, with a language model judging which pages describe digital technologies. Its 184 base technologies, the finest level, are grouped into intermediate technologies and these into root domains. The green technologies come from the OECD ENV-TECH classification, which assigns patents to environment-related technologies through their classification codes (Haščič and Migotto, 2015). Exposure is the citation-weighted semantic similarity between patent text and task or activity text, computed from sentence embeddings, which represent each text as a vector so that texts of similar meaning lie close together. It builds on the approach of Prytkova et al. (2024) and uses the same text, embeddings and aggregation in both domains.

For the digital taxonomy, about 70 percent of the pages lie closer to the centre of their own technology than to any other. In the median base technology, patents of the same technology share patent classes more often, and cite one another about thirty times as often, as patents of different technologies, although the construction never uses either kind of information. The candidate pages match 94 percent of the patent corpus, and the unmatched patents form no large citation-connected group. We compare the exposure measure with ten published exposure indices. Exposure to the artificial-intelligence root correlates positively with the five AI and generative-AI indices, and exposure to the robotics root correlates negatively with them and positively with a robot index. The construction of the measure requires correction in 1.5 percent of the exposed digital cells, the pairs of an occupation or industry and a technology, and in 0.07 percent of the exposed green cells. The difference arises in the attribution of patents to digital technologies through their text.

Direction is set mainly by the technology and the occupation. Together they account for about four fifths of the variance of direction across the rated combinations, and across the industries in which a technology meets the same occupation the direction almost never changes sign. The ratings

are stable when the model is queried repeatedly, with split-half correlations above 0.9. Substitution is rated above complementarity in only about one combination in ten, and the two constructs rise together, so that direction is the comparatively narrow gap between two intensities of the same contact. The level of this gap is partly a property of the model (Section 5.3). The measure is associated with employment change. Across country-by-occupation cells in the EU Labour Force Survey (LFS) microdata, occupations with more digital complementarity content grew faster over 2011–2023 than comparable occupations, and those with more digital substitution content grew more slowly. Green substitution content has a negative coefficient that is significant only in the public Eurostat table. The contents are measured on the patents filed in the years of each outcome period.

The paper's contribution is a direct measure of the direction of technological change, rated as two separate constructs by technology, occupation and industry for the digital and the green domain, together with evidence that the direction it assigns is reflected in employment. Existing measures infer direction from the spread of exposure across an occupation's tasks (Gmyrek et al., 2023) or from patent text (Autor et al., 2024; Kogan et al., 2023). Model-rated technology effects at the task level are used by Eloundou, Manning, Mishkin, and Rock (2024) and, distinguishing core from supplemental tasks, by Eisfeldt, Schubert, and Zhang (2023), and Teutloff, Einsiedler, Kässi, Braesemann, Mishkin, and del Rio-Chanona (2025) classify skills as substitutes for or complements to generative artificial intelligence in freelancer demand. We ask for substitution and complementarity directly and separately, by technology and industry, from the employer's side as well as the worker's. The agreement between the model and experts, with its limits, is established in Rony and Ciarli (2026). This paper tests whether the model's judgements carry information about employment. The measure rests on two inputs that we construct and release, a taxonomy of digital technologies and exposure scores for both domains, with every correction recorded.

Patent classifications such as the Cooperative Patent Classification (CPC) and the International Patent Classification (IPC) are designed for examiners' searches of prior art, and their classes correspond only loosely to the technology concepts of the labour-market literature. Rubric- and expert-based alternatives score task suitability for machine learning (Brynjolfsson et al., 2018), hand-classify occupations and extrapolate by machine learning (Frey and Osborne, 2017), or delineate technology fields through keyword searches of patent titles and abstracts (Webb, 2020). They are difficult to extend to new domains. For environmental technologies the ENV-TECH classification is a common instrument for measuring green innovation (Haščič and Migotto, 2015; Santoalha, Consoli, and Castellacci, 2021; Bianchini, Damioli, and Ghisetti, 2023). Text-based exposure measures intersect patent titles with task descriptions (Webb, 2020), link benchmark progress in artificial intelligence to occupational abilities (Felten et al., 2018; Felten, Raj, and Seamans, 2021), collect task-level exposure to large language models from human and model raters (Eloundou et al., 2024), identify automation patents from patent full text (Mann and Püttmann, 2023), build patent-based exposure to artificial intelligence and robots (Gathmann, Grimm, and Winkler, 2024), measure patent novelty through textual similarity (Kelly, Papanikolaou, Seru, and Taddy, 2021), or read

exposure off the labour market itself (Alekseeva, Azar, Giné, Samila, and Taska, 2021; Peede and Stops, 2024). Our exposure measure builds on Prytkova et al. (2024), who embed patent titles and task statements in a common sentence-embedding space (Reimers and Gurevych, 2019) and cluster patents into technologies. We define the technologies through a taxonomy built from Wikipedia instead and extend the measurement to the green domain on identical occupation and industry text. The taxonomy builds on the use of Wikipedia's category and link structure as a source of semantic relations (Ponzetto and Strube, 2007; Mihalcea and Csomai, 2007) and on community detection with the Leiden algorithm (Traag, Waltman, and van Eck, 2019; Blondel, Guillaume, Lambiotte, and Lefebvre, 2008; Newman, 2006), as an alternative to topic models of patent corpora (Suominen, Toivanen, and Seppänen, 2017; Grootendorst, 2022) and to direct clustering of patent embeddings (Dhillon and Modha, 2001).

Section 2 constructs the taxonomies, Section 3 validates the digital one and Section 4 constructs and validates the exposure measures. Sections 5 and 6 construct and validate the direction measure, and Section 7 concludes. The appendices give the data, the full construction and the evidence behind each design choice. Experiments that did not lead to a design choice are in a separate experiment log.

2 The technology taxonomies

The technology units must be fine enough to separate technologies that affect different work, computer vision from language processing or robot manipulators from navigation systems, and they must be defined from external data so that the construction can be repeated and extended. We use the OECD ENV-TECH classification for the green domain and construct a taxonomy from Wikipedia and patents for the digital domain.

2.1 Collecting and refining the digital pages

Starting from 27 curated seed categories, a breadth-first crawl descends the English Wikipedia category graph one level at a time to a depth of four category levels. It yields about 1.8 million records, each a page and the category path through which it was reached. The crawl is deliberately over-inclusive. A page is reached through six category routes on average, and biographies, films and geographic entries share categories with technology topics. A five-stage cleaning cascade, ordered by computational cost, reduces the records to about 78,000 pages. The page descriptions are then matched against 1.9 million digital patent abstracts in the embedding space of PatentSBERTa, a sentence-embedding model trained on patent text (Bekamiri, Hain, and Jurowetzki, 2024), and three filters reduce the matched pages to the candidate pages. These are noise filters on page titles, a patent-similarity threshold that retains pages whose best-matching patent reaches a cosine similarity of 0.6, and the LLM as judge, which classifies whether each remaining page describes an actual technology. About 8,000 pages remain. Table 1 reports the funnel, and Appendix B.1 describes each stage.

Table 1: Construction funnel of the digital taxonomy, from seed categories to the final taxonomy.

Stage	Object	Count
Seeds	Curated Wikipedia categories	27
Crawl	Raw (page, category-path) records	1,801,777
Cleaning	Pages surviving the cleaning cascade	78,049
Matching	Pages embedded and matched against patents	70,714
Filtering	Pages after the noise filters and the 0.6 threshold	11,906
Judging	Pages accepted by the LLM as judge	8,069
Clustering input	Pages after removing listed noise titles	8,011
Corrections	Pages after the correction ledger	7,398
Review, first round	Pages after the page review	7,351
Review, second round	Pages in the final taxonomy	6,928
Final taxonomy	Root / intermediate / base technologies	10 / 43 / 184

2.2 Structuring the pages into a taxonomy

The pages are grouped by community detection on the wiki-link graph, not by clustering their embeddings, because editorial cross-references can keep a specialised page within its domain even where its embedding lies far from any cluster centre (Appendix B.5 compares direct clustering). Community detection partitions a graph into groups of nodes that are more densely connected to one another than to the rest of the graph. Here the graph has one node per page and an edge wherever one page links to another. Each edge (u, v) is weighted by the similarity of the two page descriptions in the PatentSBERTa space,

$$w_{uv} \;=\; \max\{0,\ \cos(e_u, e_v)\}, \tag{1}$$

where e_u is the embedding of page u's description, and edges with $w_{uv} < 0.20$ are removed, about half a percent of the edges. Pages in connected components of fewer than five pages, some 500, are set aside as peripheral, and the remaining core pages are clustered.

The hierarchy is built from the top by the Leiden algorithm (Traag et al., 2019), which maximises the resolution-weighted modularity

$$Q_\gamma \;=\; \frac{1}{2m} \sum_{u,v} \left(w_{uv} - \gamma \, \frac{k_u k_v}{2m} \right) \mathbf{1}[c_u = c_v], \tag{2}$$

where k_u is the weighted degree of page u, m the total edge weight, c_u the community of u and γ the resolution, a larger γ yielding more and smaller communities. Leiden first partitions the core graph into root domains, then the subgraph of each root into intermediate clusters and the subgraph of each intermediate cluster into base technologies. The partition is adaptive in that the resolution is chosen separately for every group that is split, so that the number and size of its children suit the level. The target number of children is 8 to 12 at the root, 5 to 8 at the intermediate and 2 to 6 at the base level, with minimum child sizes of 500, 25 and 8 pages, and a group is split only if it

holds more than 3,000, 100 or 30 pages and at least twice the minimum child size. For each group a binary search over γ finds the range in which Leiden returns a number of children K_γ within the target $[K_{\min}, K_{\max}]$, fifteen values of γ are taken evenly from that range, and the value minimising

$$\text{score}(\gamma) \;=\; \mathbf{1}\big[K_\gamma \notin [K_{\min}, K_{\max}]\big] \;+\; \lambda_s\, u_\gamma \;-\; \lambda_m \left(\bar{w}_\gamma^{\,\text{in}} - \bar{w}_\gamma^{\,\text{bt}}\right) \tag{3}$$

is retained, where u_γ is the share of children smaller than the minimum size, $\bar{w}_\gamma^{\,\text{in}}$ the mean similarity of pages to the centre of their own child, the mean of the child's page embeddings, and $\bar{w}_\gamma^{\,\text{bt}}$ the mean similarity between the centres of the children, with $\lambda_s = 1$ and a margin weight $\lambda_m = 8$, the margin being the within-child less the between-centre similarity. The score selects the resolution at which the children are most distinct in content among those within the level's size range.

Two corrections follow the top-down pass. Branches with a single child are collapsed into their parent, and every intermediate or base cluster that holds more than 90 percent of its parent's pages, or whose pages are on average closer to the centre of another cluster than to their own, is split once more. Its subgraph is pruned at a local threshold of 0.25 at the intermediate and 0.30 at the base level, the resolution search is rerun with the upper bound on the number of children raised to sixteen and twelve, children of a single page are marked peripheral, and the new split is kept if it raises the mean margin of the cluster's pages or the share of its pages closer to their own centre than to any other. Each peripheral page then joins the base cluster to which its wiki-links carry the largest total edge weight, and pages without a link to the core remain unassigned. The free parameters, the weight of semantic similarity against link counts, the pruning threshold and the margin weight, were selected by a grid search on the within-minus-between margin at the worst of the three levels (Appendix B.4). Clustering the patent embeddings directly would assign every patent, but it yields unlabelled partitions that omit documented technologies without separable patent clusters (Appendix B.5).

The raw clustering, 12 root, 56 intermediate and 205 base clusters, becomes the final taxonomy in two steps. A correction ledger records the structural and page-level corrections, some 1,800 in a first pass and, in a second, the split of 30 base clusters into 62 and some 1,100 page moves. The ledger places clusters and pages within the hierarchy and removes some 600 pages (Appendix B.8). A language-model review then reads every technology with its pages and judges, on technology content alone, whether each page belongs to its technology, to another technology or to none. The review, whose proposals the authors adopted after inspection, removes pages that are not digital technologies, moves pages to the technologies they describe, reduces duplicate pages to one, creates technologies for coherent groups of pages that the clustering had placed with others, such as service robots, wearable devices and distributed ledgers, and separates a root for transport and navigation systems from the robotics root (Appendix B.7). Table 2 lists the ten root domains of the final taxonomy, with 43 intermediate and 184 base technologies on about 6,900 pages.

Table 2: Root domains of the final digital taxonomy.

Root domain	Intermediate clusters	Base technologies	Pages
Computing Hardware and Systems	3	15	1,333
Software and Data Management	8	45	1,136
Networking and Telecommunications	4	12	954
Signal Processing and Communications	7	32	914
Artificial Intelligence	4	10	791
Imaging and Display Technology	2	13	566
Cryptography and Security	2	11	474
Robotics and Manufacturing	5	24	455
Transport and Navigation Systems	3	6	201
Quantum Computing	5	16	104
Total	43	184	6,928

2.3 The green taxonomy

For the green domain we adopt the existing OECD ENV-TECH classification (Haščič and Migotto, 2015). Its 66 technology nodes in ten groups are defined by mappings from the CPC codes that patent examiners assign, so that a patent's membership in a node follows from its classification codes alone (Appendix B.10).

3 Validity of the taxonomy

A taxonomy built from embeddings could reproduce the geometry of the embedding space without corresponding to technologies, so we assess the digital taxonomy on two kinds of evidence. Internal validity asks whether, in the space from which the taxonomy was built, the pages of a technology lie close to one another and apart from other technologies, and whether the partition is stable when the clustering is rerun. External validity rests on the patent classes that examiners assign and the citations that inventors make, information that the construction never uses, and on the coverage of the patent corpus that the taxonomy is meant to classify.

3.1 Internal validity

Figure 1 reports four checks at each level of the final taxonomy. Semantic coherence, the mean over technologies of the mean similarity of a page to its technology's centre, rises from 0.58 at the root to 0.73 at the base level, and the margin share, the share of pages closer to the centre of their own technology than to that of any other of the same level, is about 70 percent at each level. Before its final resplitting round, the clustering run from which the taxonomy is derived attains positive within-minus-between margins at all three levels, and its partition is stable across reruns. Over three random seeds the adjusted Rand indices, a chance-corrected measure of agreement between two partitions, lie between 0.69 and 0.88. The margin of an individual technology, the mean

similarity of its pages to its centre less the similarity of that centre to the nearest other centre of the same level, is negative for every root and intermediate technology and for about two thirds of the base technologies. In the raw clustering the margins fall with cluster size, so that the negative margins concentrate in the largest clusters (Appendix B.6).

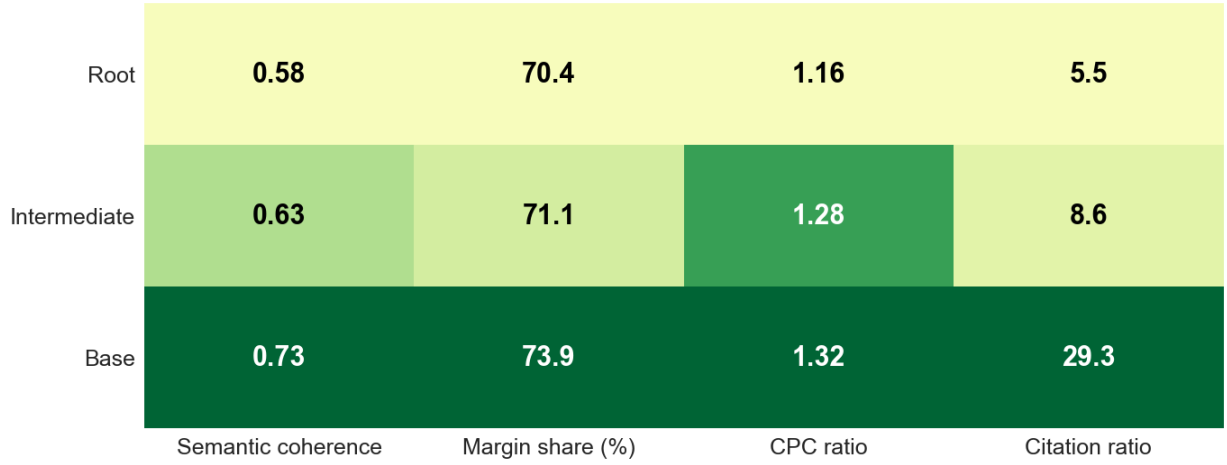


Figure 1: Validation of the final digital taxonomy. The columns report four measures at the root, intermediate and base levels. Semantic coherence is the mean similarity of a page to its own cluster centre, and the margin share is the share of pages closer to their own cluster centre than to any other. The CPC ratio and the citation ratio are computed for patents assigned to technologies through their best-matching page, as the median over technologies of the within-technology relative to the between-technology rate of shared patent classes and of citations per pair of patents. Coherence and the two ratios rise with depth, and the base-level citation ratio is 29.3.

3.2 External validity

The two external criteria in Figure 1 assign each patent of the corpus used for exposure to the technology of its best-matching page, at a similarity of at least 0.5, and compare, per pair of patents, how often patents of the same technology share three-digit patent classes and cite one another with how often patents of different technologies do. The CPC ratio and the citation ratio are the medians of these two ratios over the technologies of a level. The CPC ratio rises from 1.16 at the root to 1.32 at the base level, and the citation ratio from 5.5 to 29.3, so that in the median base technology patents of the same technology cite one another, per pair, about thirty times as often as patents of different technologies.

Coverage is assessed before clustering, on the 8,069 candidate pages and the 1.9 million patents of the matching corpus, through three channels, the embedding similarity that drives the matching, the CPC classes that patent examiners assign and the citations that inventors choose. Figure 2 displays the distribution over patents of the similarity to the closest page, and 94 percent of the patents reach the threshold of 0.5. Pairs of patents matched to the same page have three-digit CPC sets about 2.7 times as similar as pairs matched to different pages, and cite one another about

57 times as often. The 6 percent of patents below the threshold are dispersed. Among them, the citation-connected components of at least ten patents, groups linked to one another by chains of citations, hold some 7,000 patents, the largest about 3,200. A technology the taxonomy missed would not be missing at random, because its patents cite one another and would surface as a large connected component. The CPC codes over-represented among the uncovered patents belong to non-digital fields such as fertilisers and lubricants. The same diagnostic decided the patent text on which the matching relies. With patent titles instead of abstracts, a single component of almost 380,000 patents holds almost half of the uncovered patents (Appendix B.2).

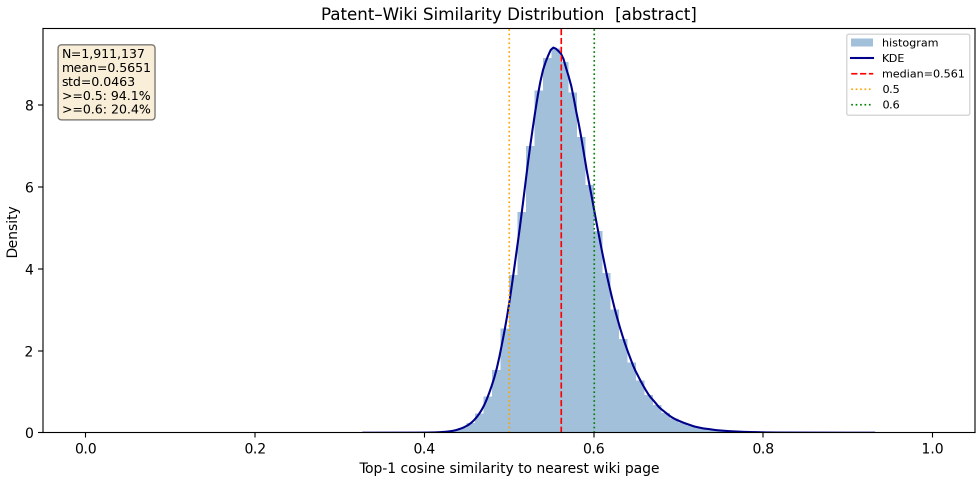


Figure 2: Distribution of the top-1 similarity, the cosine similarity of each patent to its closest page. Patent abstracts are matched against page descriptions, for 1,911,137 patents and 8,069 pages. The distribution is centred at 0.565, with 94.11 percent of patents at or above the 0.5 threshold, which lies in its lower tail. Reducing the threshold to 0.45 adds 5.7 percentage points of coverage, while raising it to 0.55 removes 33.7 percentage points.

The structure has imperfections. One intermediate cluster, Quantum Error Correction, has a single child. The largest intermediate cluster, Computer Hardware and Peripherals, spans some 650 pages in eight base technologies, and the Quantum Computing root, about a hundred pages in sixteen base technologies, contains base technologies of one or two pages. The largest base technologies, Network Protocols and Infrastructure, Computer Hardware, Peripherals and Storage, and Processor and Memory Architecture, hold several hundred pages each and were not subdivided further. The manual input to the construction is recorded in the correction ledger and in the record of the language-model review, which list every structural decision on named clusters and pages (Appendices B.8 and B.7).

4 Exposure

Exposure measures how strongly each technology bears on the work of each occupation and the activities of each industry. The measure reads this from text, comparing what the technology's

patents describe with what the occupation's tasks or the industry's activities describe, so that it can be computed identically for every technology of both domains.

4.1 Measuring exposure

For a technology n and an entity e, an occupation or an industry, exposure measures technological relevance, the degree to which the content of the technology's patents resembles the tasks of the occupation or the activities of the industry. The measure is neutral with respect to direction by design. The bank-teller example of the introduction shows that the sign of a technology's employment effect is not a property of the technology alone, so exposure delivers the magnitude of contact and leaves the direction to the separately rated measure of Section 5. Occupations are 427 of the 436 four-digit unit groups of the International Standard Classification of Occupations (ISCO-08), represented by some 2,800 task sentences from the European Skills, Competences, Qualifications and Occupations classification (ESCO), and industries are the 272 three-digit groups of the Statistical Classification of Economic Activities in the European Community (NACE Rev 2), represented by some 850 activity sentences (Appendix A).

The construction builds on Prytkova et al. (2024), who relate patent and task text, and connects three corpora, the entities' task or activity sentences, the patents and the taxonomy pages, through three similarity relations with distinct roles. Membership determines which patents constitute a technology, value measures how strongly a patent resembles an entity's tasks or activities, and gating determines whether a sentence relates to the technology's subject at all. Because the three relations are separate, their errors can be corrected separately (Section 4.2).

Membership operates in the PatentSBERTa space in which pages and patents were matched for the taxonomy. Each patent is tagged to the pages among its twenty most similar with a cosine similarity of at least 0.40, and it counts towards a single technology, the one that contains its best-matching page. Under a many-to-many rule, in which a patent joins every technology in which it has a qualifying page, a small number of broad technologies absorbs a disproportionate share of the corpus. The value of a patent for an entity sentence is their cosine similarity in the space of the general-purpose sentence-embedding model all-mpnet-base-v2 (MPNet below) (Song, Tan, Qin, Lu, and Liu, 2020; Reimers and Gurevych, 2019), counted only above a floor of 0.32, the 99th percentile of the sentence-patent similarities. The floor is defined on the percentile scale because similarity scales differ by about a factor of two between the two embedding models and differ across the three relations within a model, so that an absolute threshold selects tails of very different depth (Appendix C.3). A sentence contributes to a technology through a patent only if its PatentSBERTa similarity to the page through which the patent belongs to the technology reaches 0.40. This page gate prevents value from accruing through patents whose page is unrelated to the entity's work. Each patent may give value to at most the ten entities whose sentences resemble it most once each entity's baseline similarity to all patents is subtracted, a variant of the top-ten relevance filter of Prytkova et al. (2024), so that no single generic patent contributes to every entity.

Figure 3 reports, over 18.7 million random triples of a task, a patent and a page, the mean task-

page similarity within bins of the task-patent and the patent-page similarity. It is highest where both are strong. Part of this agreement is geometric, since a task close to a patent that is close to a page is itself partly close to the page. The figure shows that the three relations are consistent, a precondition for requiring all three jointly. The joint requirement removes links that are strong in one relation only, such as a patent that resembles a task through shared vocabulary while belonging to an unrelated page.

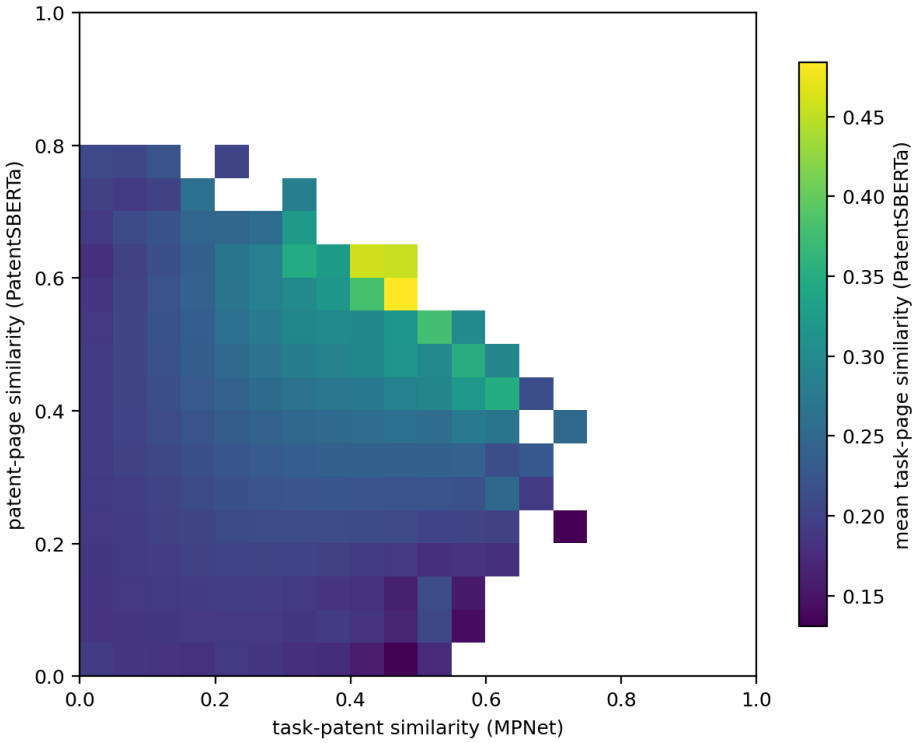


Figure 3: Agreement of the three similarity relations of the exposure measure. The sample is 20 million randomly drawn triples of a task sentence, a patent and a page of the taxonomy before the language-model review, of which the 18.7 million with non-negative task-patent and patent-page similarity are binned. Each cell reports the mean task-page similarity (PatentSBERTa, the gating relation) within a bin of the task-patent similarity (MPNet, the value relation, horizontal) and the patent-page similarity (PatentSBERTa, the membership relation, vertical). Empty cells hold no triples. The task-page similarity is highest where the other two relations are jointly strong, 0.37 in the bins in which both exceed 0.5 against 0.20 over all triples.

Let $S(e)$ denote entity e's sentences, $P_n(w)$ the member patents of technology n filed in window w, ν_p the citation weight of patent p, its forward citations relative to the mean of its filing-year cohort, so that an uncited patent carries no weight, $A(s,p)$ the value-relation cosine similarity, and $g(s,p)$ the indicator that the pair passes the floor, the page gate and the relevance filter. The exposure of entity e to technology n over window w is

$$x_{en}(w) \;=\; \operatorname{asinh}\left(\frac{1}{|S(e)|}\sum_{s\in S(e)}\sum_{p\in P_n(w)}\nu_p\cdot\max\{0,\,A(s,p)\}\cdot g(s,p)\right), \tag{4}$$

the mean over the entity's sentences of the citation-weighted sum over the technology's patents, rescaled once by the inverse hyperbolic sine. The mean over sentences rewards technologies whose patents relate to many of the entity's tasks, while a maximum would reward a single task. Appendix C.4 lists the alternative aggregations. Within a sentence the sum runs over all member patents that pass the gates, so exposure grows with the number of such patents. The measure is computed year by year, with each patent weighted by its citations relative to the mean of its filing-year cohort (Hall, Jaffe, and Trajtenberg, 2001), and one untransformed value is stored per year, so that exposure over any window is the inverse hyperbolic sine of the summed yearly values and the nonlinearity is applied exactly once. Because an uncited patent carries no weight, the weight of the most recent cohorts rests on the minority of their patents that have already been cited (Appendix C.6). An entity's exposure to a domain is the mean over the domain's N technologies, $X_e(w) = N^{-1}\sum_n x_{en}(w)$, with technologies that do not reach the entity counted as zero. Exposure to a digital root or intermediate technology, or to a green group, is the mean of $x_{en}(w)$ over the base technologies it contains, with zeros included, the rule of the domain average X_e applied within the root or group. The green measure follows the same construction except in membership, which the ENV-TECH indicators fix and which needs no page gate (Appendix C.1).

4.2 The LLM as judge

The last step of the construction uses the LLM as judge. It reads the entity rankings and the largest patent links, the contributions of single patents to an entity's exposure, and sets to zero those without technological content. It applies two criteria. The top-ranked entities of a technology must be plausible users or producers of it, and no ranking may be driven by a lexical coincidence. In the digital domain it reviews up to the fifty largest patent links of every exposed technology, for occupations and for industries. Its verdicts on some 15,000 links move almost 4,000 patents to the technology to which they belong, remove about 300 that are not digital technologies and set some 6,400 links to zero. It also sets to zero the entity-technology cells driven by a lexical coincidence, such as refuse sorters ranked on a data-structures node through the word sorting, 1.5 percent of the exposed digital cells. In the green domain, where membership is fixed by classification codes, it sets to zero 0.07 percent of the exposed cells, those of two occupations of drivers of animal-drawn and hand-propelled vehicles and one industry, the manufacture of transport equipment not elsewhere classified, whose description names hand-propelled and animal-drawn vehicles, ranked on the transport nodes through the word vehicle. No technology is dropped for lack of exposure. The digital and the green measures share the value relation, the entity text and the aggregation. They differ in the patent corpus and in how patents are attributed to technologies, through their text against pages with a page gate or through their classification codes, so the higher digital share is consistent with noisier text-based attribution, and because only the digital measure also passes

through the link review (Appendix C.2), the two sources of the difference cannot be separated.

4.3 Validity of the exposure measure

For external validity, the occupation-side exposure profiles are compared with ten published exposure indices over 356 four-digit ISCO occupations, with exposure over 2009–2023. Indices keyed to the US Standard Occupational Classification are mapped to ISCO-08 through a crosswalk between SOC-2010 and ISCO-08 derived from the Bureau of Labor Statistics, averaging over the SOC codes that map to one ISCO code, and the 356 occupations are those with a value in every index after the mapping. The ten indices are the AI occupational exposure index of Felten et al. (2021), the model- and human-rated language-model exposure of Eloundou et al. (2024), the generative-AI exposure of Eisfeldt et al. (2023), the AI applicability index of Tomlinson, Jaffe, Wang, Counts, and Suri (2025), the computerisation probability of Frey and Osborne (2017), the AI, software and robot indices of Webb (2020), and the aggregate technology exposure of Prytkova et al. (2024). Figure 4 reports the Spearman rank correlation of each of our ten root-level profiles with each index and leaves insignificant correlations blank. The Artificial Intelligence root correlates positively with all five AI and generative-AI indices. The Robotics and Manufacturing root shows the opposite pattern, correlating positively with the Webb robot and software indices and with Frey and Osborne (2017) and negatively with the five AI and generative-AI indices. The Transport and Navigation Systems root, which the review of the taxonomy separated from the robotics root, follows the robotics pattern, negative with the five AI and generative-AI indices and positive with the Webb robot, software and AI indices. The broad aggregate of Prytkova et al. (2024) correlates positively with every root but quantum computing, whose occupation profile correlates weakly and positively with four of the five AI and generative-AI indices, at 0.10 to 0.14, and with none of the other indices. The aggregate of Prytkova et al. (2024) shares the approach of our measure and one of its authors, so its correlation with the roots is partly mechanical, and the other nine indices carry the comparison. The AI index of Webb (2020) correlates only weakly with the Artificial Intelligence root, at 0.12. It correlates more strongly with the roots of physical and signal technologies, robotics and manufacturing, transport and navigation, computing hardware and signal processing, as does Webb's software index, and Webb's robot index chiefly with the first three. A size or generic-technology artefact would correlate every root with every index. The sign reversals argue against such an artefact.

Internally, the measure rests on the agreement of its three relations (Figure 3), and the measure is validated by reading its complete entity rankings, because the relevant failure modes, lexical coincidences and concentration artefacts, are not visible in correlation-based statistics. Figure 5 reports the most exposed occupations by domain. The top digital occupations are information and communication technology (ICT) managers, ICT operations and support technicians and applications programmers, together with precision-instrument makers, photographers and medical-imaging technicians. The top green occupations are chemical, power and steam plant operators, incinerator and water-treatment plant operators, air-conditioning and refrigeration mechanics, and refining opera-



Figure 4: External validity of the occupation exposure profiles of the roots. The cells report Spearman rank correlations with ten published exposure indices over 356 four-digit ISCO occupations, with exposure over 2009–2023. Blank cells are insignificant at the five-percent level. The Artificial Intelligence root correlates positively with the five AI and generative-AI indices, and the Robotics and Manufacturing and the Transport and Navigation Systems roots correlate negatively with them and positively with the robot index. The aggregate index of Prytkova et al. (2024) correlates with every root except quantum computing.

tors. At the level of individual technologies, for example, Electronic Payment Processing ranks bank tellers and cashiers among occupations and monetary intermediation among industries, and Medical Imaging ranks medical-imaging technicians and medical-device manufacturing. Geothermal energy ranks well drillers first, and wastewater treatment ranks water-treatment plant operators first. *The chemical and refining operators at the top of the green ranking are rarely counted among green occupations, and the technologies behind their rank show why. Chemical processing plant controllers, the occupation most exposed to green technologies, rank first among occupations on the technologies of oil refining and petrochemicals, on fuel cells, on hydrogen for transport and on environmental monitoring, and second on the technologies of the chemical industry, on hydrogen technology and on the capture of carbon dioxide and of methane. Petroleum and natural gas refining plant operators rank second on oil refining and petrochemicals and on hydrogen for transport and third on fuel cells and on hydrogen technology. These ENV-TECH classes describe cleaner production processes and new fuels in the industries with the largest emissions, and the occupations that run those processes are the ones the measure places at the top. Environmental innovation is most relevant to the activities with the largest environmental impact, and the ranking reflects that.*

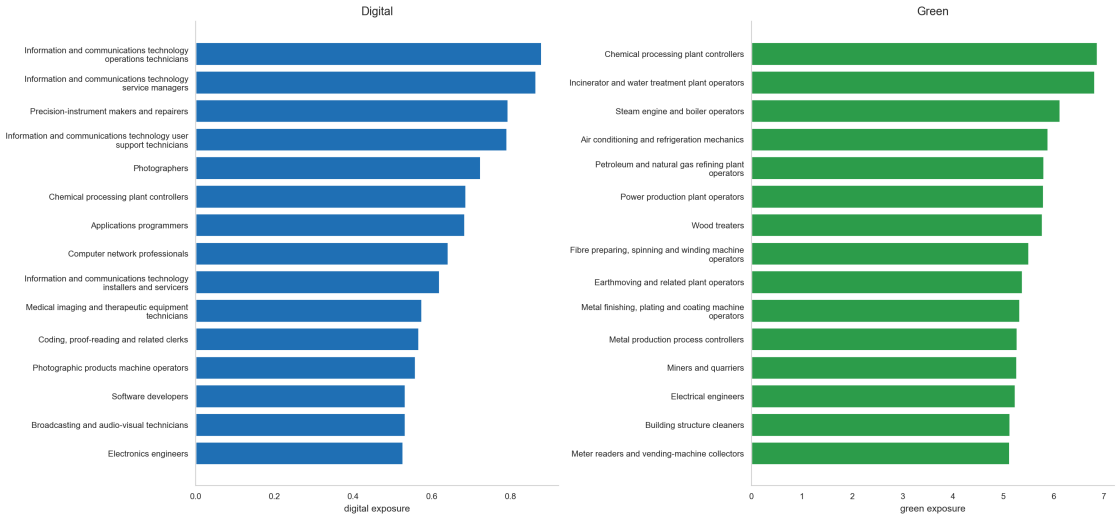


Figure 5: The most exposed occupations by domain (exposure over 2009–2023). Digital exposure is led by ICT occupations together with precision-instrument and imaging occupations, and green exposure by plant and machine operators in energy, water and process industries.

The residual occupation-side limitation is a bias towards concretely worded task statements. Task text written in operational language is systematically closer to patent text than abstract task text, so that under some aggregations abstract technologies rank manual occupations above their actual users. Appendix C.5 describes the bias.

4.4 The structure of the two exposures

Figures 6 and 7 report the median exposure over 2009–2023 by technology root or group across major occupation groups, and Figures 23 and 24 in Appendix C.8 across industry sections. Within

the digital domain the roots reach different occupations. Artificial intelligence and software reach professionals first and networking and cryptography clerical workers, whereas robotics and manufacturing, computing hardware, imaging and display, and transport and navigation reach plant and machine operators first, robotics also elementary occupations and craft trades. Across industries the information and communication section leads for seven of the ten digital roots, among them cryptography and security, for which finance and insurance follows. Construction leads for robotics and manufacturing, followed by electricity and gas, water and waste and manufacturing, and transport and storage leads for transport and navigation. The median industry of every section has zero exposure to quantum computing. Green exposure is highest for craft trades, plant and machine operators and skilled agricultural workers, and concentrates in electricity and gas and in water and waste, followed by mining, construction, manufacturing and transport.

Figure 25 in Appendix C.8 reports the correlations between the ten digital roots and the ten green groups in their exposure across four-digit occupations. The green groups other than ICT and smart grids form one block, correlated at 0.5 to 0.9 with one another, while the digital roots are weakly to moderately correlated with one another. Across domains, the artificial-intelligence, software, cryptography, networking, signal-processing, imaging and quantum roots are at most weakly correlated with the green block, whereas computing hardware and transport and navigation correlate with it moderately and robotics and manufacturing strongly, at 0.5 to 0.8. The ICT group of the green classification correlates positively with every digital root. Across three-digit occupation groups the two domain aggregates correlate at 0.45 (Figure 8). The two exposures overlap only partly, and the groups exposed to both are few, among them process control technicians, chemical-products plant operators, electronics and telecommunications installers and electrotechnology engineers. Figure 26 in the same appendix reports the distribution of each exposure across four-digit occupations. Digital exposure is concentrated at low values with a long right tail, while green exposure is spread more evenly.

5 The substitution and complementarity measure

The direction measure rests on two constructs, substitution and complementarity, rated by a large language model under workplace personas, roles that the model is asked to adopt, for the technology-occupation-industry cells in which exposure concentrates (Appendix D).

5.1 Unit and rating design

The rated unit is a triplet consisting of a technology node (a digital base technology or a green ENV-TECH node), a three-digit ISCO occupation group, and a NACE Rev 2 section, the finest occupation-industry resolution at which European employment data record the pair. The base-technology level is chosen because we expect specific technology labels to draw specific assessments, and ratings can be aggregated to coarser levels while the converse is not possible.

Two constructs are rated. Substitution is defined as the technology taking over the work itself, so

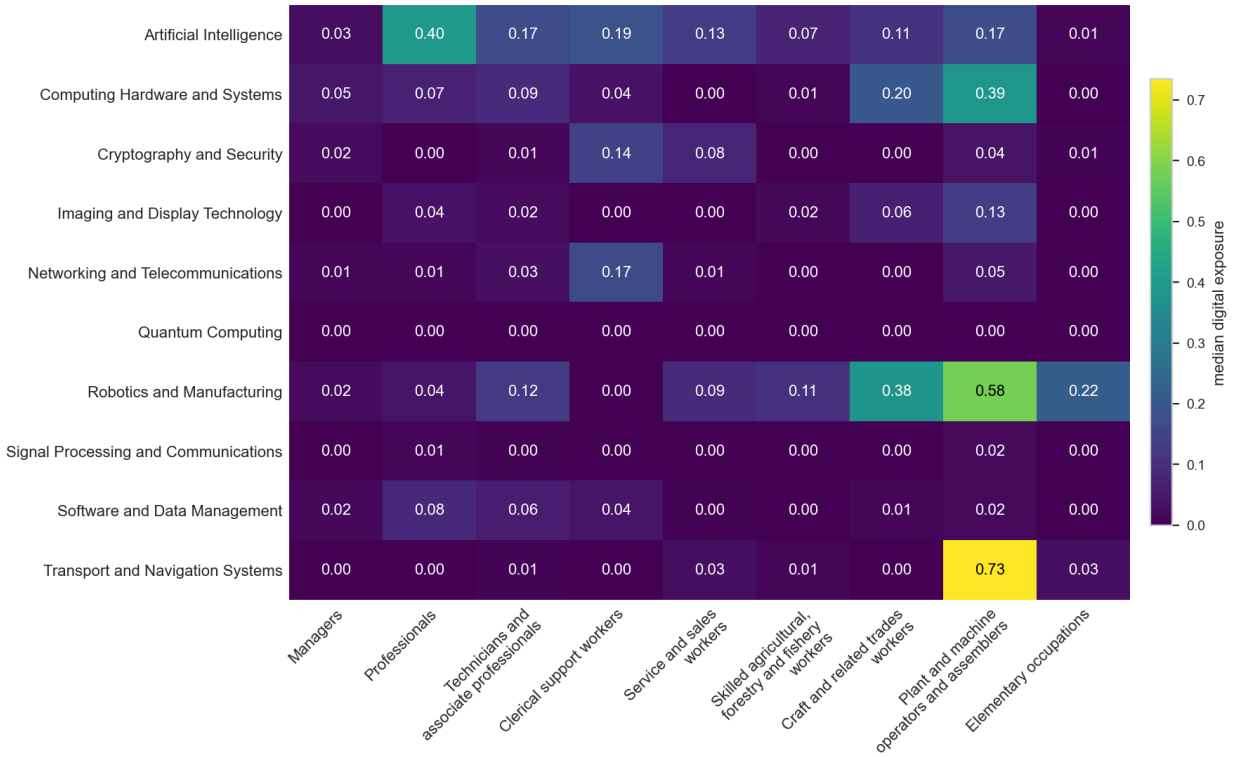


Figure 6: Median digital exposure over 2009–2023 by technology root and ISCO-08 major occupation group. Artificial intelligence and software reach professionals first, networking and cryptography clerical workers, and robotics and manufacturing, computing hardware and transport and navigation plant and machine operators.



Figure 7: Median green exposure over 2009–2023 by ENV-TECH group and ISCO-08 major occupation group. Exposure is highest among craft trades, plant and machine operators and skilled agricultural workers.

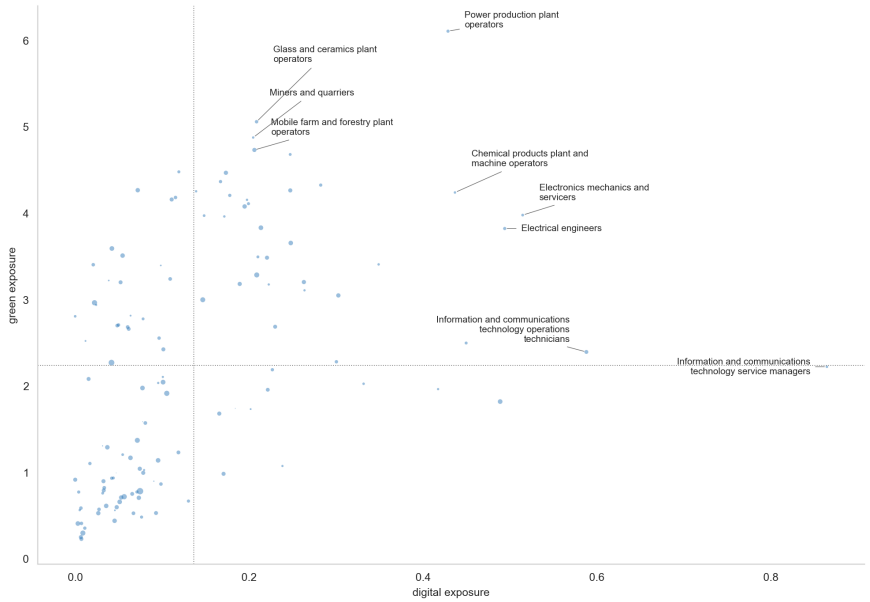


Figure 8: Digital against green exposure across 126 three-digit occupation groups. Exposure is over 2009–2023, in the units of each domain. Point size is EU employment and the grey lines mark the means. The two exposures correlate at 0.45, with a rank correlation of 0.58, and the groups exposed to both are few. The four groups highest on digital exposure, the four highest on green exposure and the three highest on both are labelled with the title of their first four-digit occupation.

that fewer workers are needed, and complementarity as the technology supporting or enhancing the workers, who continue to perform the work at higher quality or speed. Each construct is presented as an agreement statement and rated on a scale from 1 (strongly disagree) to 5 (fully agree), with 0 defined as do not know. The statements name no time horizon, because the rated technologies already exist, with patents filed within the exposure window 2008 to 2023. Appendix D.1 gives the prompts verbatim. The instrument has no separate not-applicable option, and Appendix D.5 shows that the zero is used as one. Each statement names the rated object as a digital or a green technology according to the domain of its node, and this domain label supplies the context in which the node's name is read.

Each rating is given under one of three personas, specified by a single role sentence. They are a manager who runs the team performing the work, an experienced worker who performs it, and a chief executive responsible for the organisation's strategy and investment decisions. Ratings are stored per persona. The chief executive represents the side of the employment relation that decides on adoption, since investment and staffing decisions are the channel through which an assessment of substitution reaches labour demand.

The respondent model is DeepSeek-V3.1, a model with open weights, queried as `deepseek/deepseek-chat-v3.1` through the OpenRouter interface in September and October 2026, at temperature 0.3 and top-p 0.3, settings that limit the randomness of the sampled answers, as selected in our companion work on language-model survey panels (Rony and Ciarli, 2026). The interface

routes each call to one of several hosting providers, and the provider was not recorded. Responses are constrained to a fixed machine-readable format, a JSON object holding one integer rating followed by a short justification. Each triplet receives 30 ratings, across two constructs, three personas and five independent draws. Each call, a single query to the model, carries one construct and one persona. In a test on an earlier version of the prompt, merging constructs or personas into one call moved the ratings away from those of separate calls, reduced the use of the zero category and raised the draw-to-draw noise (Appendix D.2). Scale, response format and the one-construct, one-persona call follow Rony and Ciarli (2026).

5.2 The rated grid

The grid, the set of triplets put to the respondent model, is restricted to cells in which both exposure and employment indicate that the question is meaningful. It uses the exposure of Section 4, computed over the window 2008 to 2023 and aggregated from four-digit occupations to three-digit groups and from three-digit industries to sections by unweighted means over the exposed codes. A triplet qualifies if the exposure of its occupation group to the node and the exposure of its section to the node both exceed a threshold. The thresholds are common to all technologies, with one for occupations and one for industries in each domain. A node-specific threshold would admit the upper tail of nodes whose exposure is low throughout. Each threshold is the 70th percentile of the corresponding distribution of exposed cells (Appendix D.3 gives the values). The grid was selected and rated in September 2026, before the window of the paper was fixed, on the window then in use, 2008 to 2023, and it was not re-selected afterwards. The descriptive statistics use exposure over 2009 to 2023, and the employment test builds the contents from the patents filed in the years of each outcome period.

The employment filter retains occupation-industry pairs accounting for at least 0.1 percent of the occupation's pooled 2011–2023 employment in the LFS, a floor that removes pairs with negligible employment. The grid has some 12,000 triplets, a third digital and two thirds green, on 112 digital and 62 green technologies. Appendix A reports its dimensions. Entities enter the prompt under their names and green nodes under their ENV-TECH descriptions, since a bare classification code does not tell the respondent what the technology is.

5.3 The ratings

The runs yield 30 usable ratings for every triplet, some 360,000 in all. The few calls that failed or returned no rating were rerun. Table 3 and Figure 9 report their distribution.

Complementarity is rated above substitution in both domains. The respondent model rates substitution about a quarter of a scale point lower than experts do (Rony and Ciarli, 2026), so the level of this gap is partly a property of the model. The analysis below relies on its variation across technologies, occupations and industries. Mean complementarity exceeds mean substitution by more than one scale point in both domains, and at the triplet level, averaging the fifteen ratings per construct, substitution is rated above complementarity in about one triplet in ten. The two

Table 3: Ratings by domain and construct. The scale runs from 0 to 5, and the zero category is counted as its bottom.

Domain	Construct	Mean	SD	n
Digital	Complementarity	3.472	1.178	60,720
Digital	Substitution	2.223	0.869	60,720
Green	Complementarity	2.965	1.357	118,935
Green	Substitution	1.934	0.911	118,935

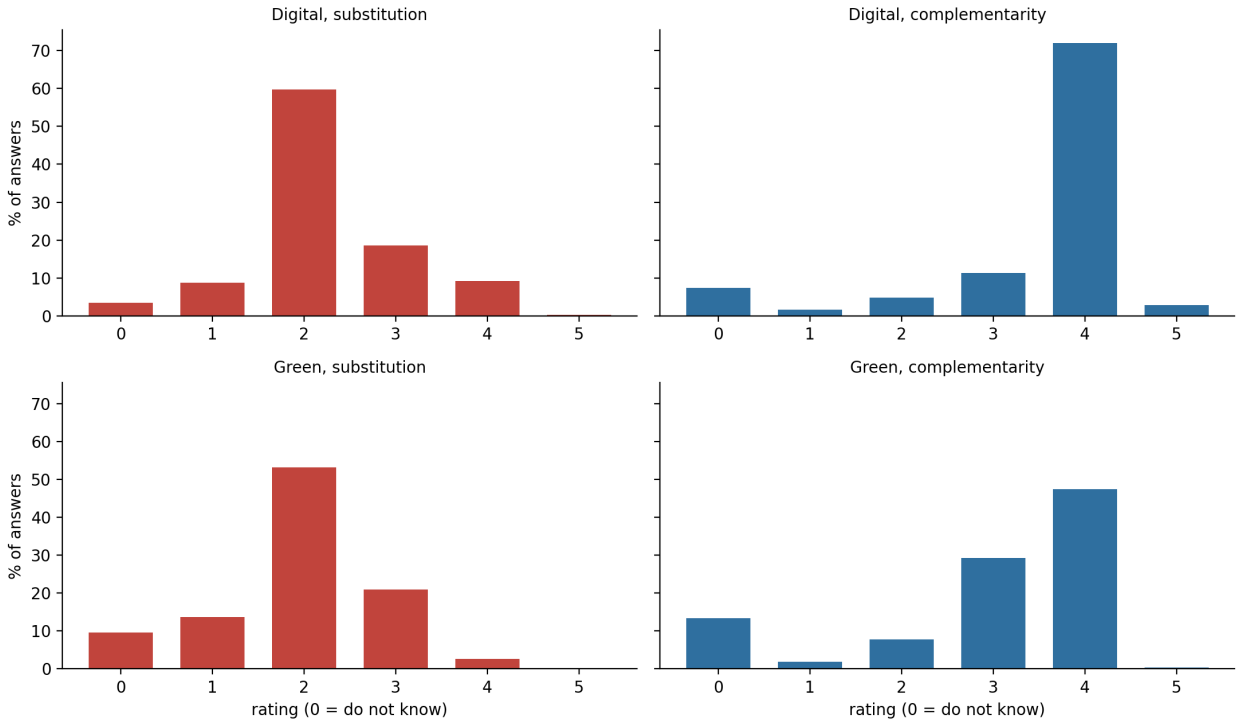


Figure 9: Distribution of the ratings by domain and construct. Substitution is modal at 2 and complementarity at 4 in both domains, the zero category is used most for green complementarity, and the top of the scale draws at most 3 percent of the ratings of any domain and construct.

constructs move together across triplets, with a Pearson correlation of 0.69 in the digital and 0.83 in the green domain (Figure 10). Both rise where the technology affects the occupation's work, and direction is the gap between them. Section 5.4 keeps both as levels. The personas move the level of the ratings and leave their ranking, and the zero category functions as a no-effect code (Appendix D).

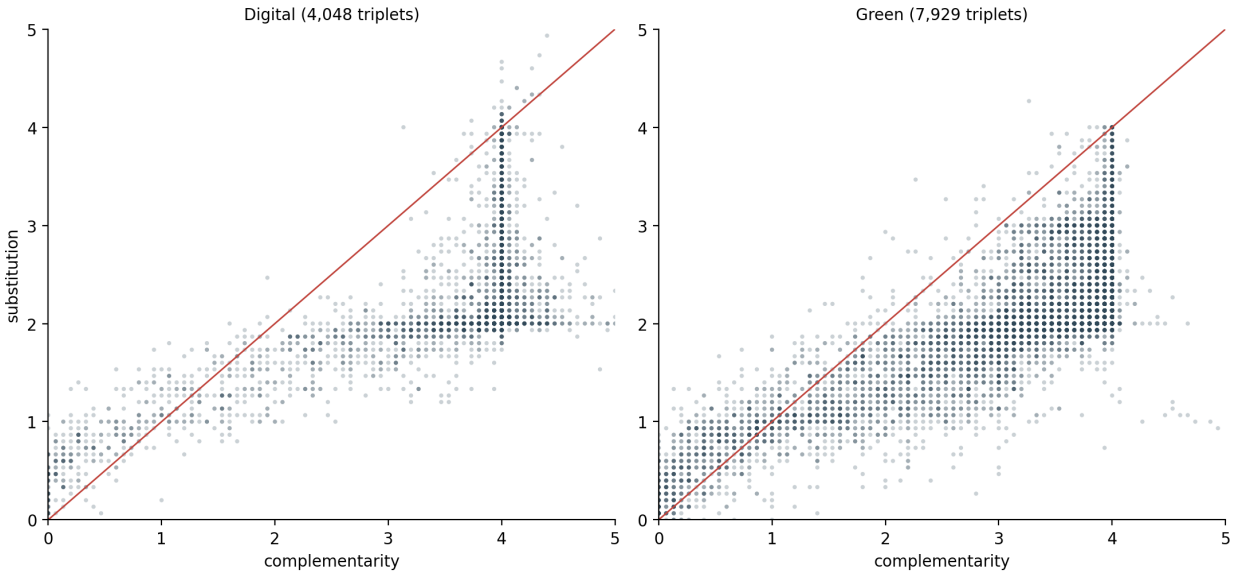


Figure 10: Triplet-level substitution against complementarity. Each point is a rated cell, and each coordinate is a mean over fifteen ratings. About 90 percent of the cells lie below the diagonal, and substitution rises with complementarity. Substitution exceeds complementarity in 354 of 4,048 digital and 795 of 7,929 green triplets.

5.4 From ratings to substitution and complementarity content

The ratings become a measure by reading each answer as a fraction of the task content the technology reaches and multiplying that fraction by the exposure. The three agreeing grades count as half, three quarters and all of the reached content, following their scale labels. The disagreeing grades and the zero count as none, so that a disagreeing answer never adds content, although the label of the higher disagreeing grade names a quarter. Appendix D.5 shows the zero and the lowest grade to be the same answer. Averaged over the fifteen ratings of a triplet the mapping gives a substitution fraction $\sigma_{n,o,i}$ and a complementarity fraction $\kappa_{n,o,i}$ for technology n, occupation group o and section i. Across the triplets σ is zero in about half and averages below a fifth, and κ averages above one half and is zero in a tenth or fewer. The mapping lowers the Spearman rank correlation between the two constructs from 0.62 and 0.77 on the raw scale to about one half between the two fractions, and σ exceeds κ in about one triplet in a hundred.

An entity's content is its exposure to each technology times the rated fraction, summed over technologies. For a four-digit occupation e in group $o(e)$, with $x_{en}(w)$ its exposure to technology n

over the window w of the application (Equation (4)) and N the number of technology nodes of the domain,

$$\Sigma_e \;=\; \frac{1}{N}\sum_n \sum_i \omega_{o(e)i}\; x_{en}(w)\, \sigma_{n,o(e),i}, \qquad \Gamma_e \;=\; \frac{1}{N}\sum_n \sum_i \omega_{o(e)i}\; x_{en}(w)\, \kappa_{n,o(e),i}, \tag{5}$$

where $\omega_{o(e)i}$ is the pooled employment share of section i among the group's workers in the LFS, and cells outside the grid contribute nothing, so that a technology whose exposure to the pair fell below the thresholds is counted as reaching no content worth rating. The industry-side content of a NACE code is the mirror image. Its exposure to each technology is multiplied by the fraction rated for its section and averaged over the occupations employed in the section, weighted by their employment shares. Σ_e corresponds to the displacement of workers from tasks in the framework of Acemoglu and Restrepo (2019), and Γ_e to the augmentation of work that stays with workers, the distinction that Autor et al. (2024) draw for patents. The rating supplies the direction and the exposure the magnitude. Because the two constructs are rated separately, σ and κ are separate intensities. For robot manipulators both are high, and $\Sigma_e + \Gamma_e$ need not equal the exposure and can exceed it, up to twice its value. The contents are also computed at the level at which exposure is measured, so they aggregate along the same paths as the exposure they accompany, to sections and industry groups on the industry side and to occupation groups on the occupation side, and can take the place of exposure in specifications that use it.

Table 4 ranks technologies by the substitution share of their assessed effect, $\bar{\sigma}_n/(\bar{\sigma}_n + \bar{\kappa}_n)$ over the technology's rated triplets. The digital ranking is led by robotics and artificial-intelligence technologies, service robotics, robot manipulators, autonomous farm equipment, symbolic AI and agent systems and language models, and ends with infrastructure and interface technologies, messaging middleware, operating-system kernels, graphical user interfaces, immersive technologies and display panels. The green ranking runs from the production-process technologies of cement, metals, chemicals and glass to energy efficiency in buildings, water conservation, health protection and conventional vehicles. The appearance of cement and glass production near the top reflects the labelling problem of Appendix D.5, in which these nodes are rated as industrial processes and are not read as mitigation technologies. A share of two small fractions is unstable. Table 4 therefore reports both fractions, and glass and cement production rank high on fractions below a quarter. Across technologies the share has a standard deviation of 0.13 in both domains, and within a technology, across the sections in which it was rated, about half that or less.

Substitution and complementarity content are correlated. Across the 427 occupations the correlation is 0.67 for digital and 0.89 for green technologies, and across the 272 NACE codes it is 0.68 and 0.94. Both contents also correlate with exposure, at 0.71 to 0.88. Because the two contents enter the employment regression of Section 6.3 jointly, the green coefficients rest on little independent variation. Digital substitution content is zero for 19 percent of occupations, mostly those with no rated cell.

Table 4: Technologies ranked by the substitution share of their assessed effect. The share is $\bar{\sigma}_n/(\bar{\sigma}_n+\bar{\kappa}_n)$ over the technology's rated triplets, and the table lists the top and bottom five per domain among technologies with at least twenty rated triplets.

Technology	$\bar{\sigma}_n$	$\bar{\kappa}_n$	Share	Triplets
Digital, highest				
Service Robotics	0.65	0.74	0.47	70
Robot Manipulators and Arm Systems	0.59	0.72	0.45	82
Autonomous Farm Equipment	0.23	0.29	0.44	26
Symbolic AI, Knowledge and Agent Systems	0.51	0.74	0.41	83
NLP, Generative AI and Language Models	0.51	0.76	0.40	92
Digital, lowest				
Messaging Middleware	0.00	0.68	0.00	24
OS Kernel, Process and Memory Management	0.00	0.49	0.01	32
Graphical User Interfaces and UX Design	0.02	0.85	0.02	23
AR/VR and Immersive Technologies	0.03	0.72	0.04	55
Display Panels	0.03	0.70	0.04	37
Green, highest				
Production of cement	0.16	0.22	0.43	22
Technologies in the production process for final industrial or consumer products	0.41	0.58	0.42	322
Technologies related to metal processing	0.36	0.51	0.41	124
Technologies relating to chemical industry	0.37	0.54	0.41	68
Glass production	0.07	0.12	0.38	43
Green, lowest				
Conventional vehicles (based on internal combustion engine)	0.00	0.23	0.00	224
In human health protection, e.g. against extreme weather	0.00	0.53	0.00	214
Water conservation; Efficient water supply; Efficient water use	0.01	0.59	0.01	197
Energy efficiency in buildings	0.00	0.44	0.01	409
Fuel efficiency-improving vehicle design	0.00	0.34	0.01	38

6 Validation of the direction measure

6.1 Stability and determinants of direction

For each triplet the direction is the difference between the complementarity and the substitution fraction, $\kappa_{n,o,i} - \sigma_{n,o,i}$. It is positive in about nine triplets in ten. When each of the fifteen ratings is assigned at random to one of two halves, the directions computed from the two halves correlate above 0.9 in both domains, a split-half stability of about 0.95 by the Spearman–Brown formula (Table 5). At the low sampling temperature the draws vary little by design, so the statistic captures the consistency of the sampled ratings and says nothing about agreement between independent raters. The rating noise in a triplet's direction has a standard deviation of 0.05. Table 5 decomposes the variance of $\kappa - \sigma$ across triplets. The technology alone accounts for the largest share in the digital domain, the technology and the occupation for similar shares in the green domain, and the industry alone for 5 percent or less. The combination of technology and occupation accounts for about four fifths of the variance, 0.81 and 0.78. Pairs rated in a single industry are fitted exactly by the pair effects, but they hold only 316 of the digital and 448 of the green triplets, and on the pairs rated in two or more industries the share is the same, 0.81 and 0.77. Within those pairs the industry and the rating noise together account for 19 and 23 percent of the variance, so the industry in which a technology meets an occupation accounts for at most about a fifth. Among the roughly 1,000 digital and 1,600 green technology-occupation pairs rated in two or more industries, the direction varies across industries by a median range of 0.17 and 0.20, three to four times the rating noise, and the range exceeds 0.2 in 41 and 50 percent of the pairs, but the direction changes sign beyond two noise standard deviations in one pair in five hundred or fewer. The largest differences run from net substitution or no relation in one industry to strong complementarity in another, as for electronic payment processing among keyboard operators, negative in information and communication and strongly positive in accommodation and food service activities.

6.2 Aggregation and the direction signal

Table 6 reports the two contents at each level of aggregation. Across NUTS-2 regions (the EU's second-level statistical regions) their standard deviation is a quarter or less of that at any other level, and across the ten industry groups from which regional values are built they correlate at 0.88 for digital and 0.91 for green. Regional values vary little and are built from ten group values, so regions cannot separate the two contents. Across three-digit occupation groups they correlate at 0.61 for digital and 0.88 for green. Appendices D.8 and D.10 compare the contents with single indices of exposure and direction and with a usage-based index.

6.3 Substitution and complementarity content and employment change

For occupation group o in country c over a period from t_0 to t_1, Table 7 estimates

$$\ln E_{co,t_1} - \ln E_{co,t_0} \;=\; \beta_\Sigma \, \Sigma_o + \beta_\Gamma \, \Gamma_o + \phi \ln E_{co,t_0} + \mu_{\text{ISCO1}} + \delta_c + \varepsilon_{co}, \tag{6}$$

Table 5: Stability and determinants of the direction $\kappa - \sigma$ across rated triplets.

	Digital	Green
A. Ratings		
Triplets	4,048	7,929
Mean direction $\kappa - \sigma$	0.459	0.381
Share of triplets with $\kappa > \sigma$	0.90	0.89
Split-half correlation	0.931	0.908
Split-half reliability (Spearman-Brown)	0.964	0.952
Standard deviation of the rating noise in $\kappa - \sigma$	0.051	0.053
B. Share of the variance of $\kappa - \sigma$ explained by fixed effects		
Technology	0.430	0.273
Occupation	0.231	0.247
Industry	0.040	0.048
Technology, occupation and industry, additive	0.585	0.496
Technology × occupation	0.812	0.780
Technology × industry	0.538	0.400
Occupation × industry	0.311	0.315
C. Technology-occupation pairs rated in two or more industries		
Pairs	999	1,586
Triplets in pairs rated in one industry	316	448
Technology × occupation, R^2 on these pairs	0.810	0.774
Share of the variance of $\kappa - \sigma$ within pairs	0.190	0.226
Standard deviation of $\kappa - \sigma$ across industries (mean over pairs)	0.103	0.110
Median range of $\kappa - \sigma$ across industries	0.167	0.200
Pairs with a range above 0.2 (percent)	40.6	49.6
Pairs changing sign beyond two noise standard deviations (percent)	0.2	0.1

Direction is $\kappa - \sigma$ per technology-occupation-industry triplet, with each fraction the mean over fifteen ratings mapped as in Section 5.4. For the split halves, the ratings of each triplet and construct are divided at random into two halves, and the noise standard deviation is half the standard deviation of the difference between the halves' directions. Panel B reports the R^2 of $\kappa - \sigma$ on the fixed effects named. Panel C covers technology-occupation pairs rated in at least two sections, and a sign change requires $\kappa - \sigma$ above twice the noise standard deviation in one section and below its negative in another.

Table 6: The two contents by level of aggregation, with exposure over 2009–2023. Units without rated content are counted at zero. Three-digit occupation groups, NACE sections and the ten industry groups are unweighted means of their four-digit occupations or three-digit industries, and regional values weight the ten group values by each region's 2009 employment shares.

		Substitution content		Complementarity content		
Level	Units	Mean	Standard deviation	Mean	Standard deviation	Correlation of the two contents
A. Digital						
Four-digit occupations	427	0.012	0.016	0.041	0.060	0.67
Three-digit occupation groups	127	0.012	0.014	0.042	0.063	0.61
Three-digit NACE industries	272	0.010	0.012	0.034	0.062	0.68
NACE sections	21	0.006	0.008	0.028	0.051	0.80
Ten industry groups	10	0.007	0.008	0.040	0.070	0.88
NUTS-2 regions	313	0.007	0.001	0.027	0.004	0.66
B. Green						
Four-digit occupations	427	0.157	0.192	0.571	0.514	0.89
Three-digit occupation groups	127	0.147	0.190	0.545	0.503	0.88
Three-digit NACE industries	272	0.218	0.176	0.701	0.497	0.94
NACE sections	21	0.151	0.202	0.566	0.625	0.97
Ten industry groups	10	0.119	0.130	0.502	0.448	0.91
NUTS-2 regions	313	0.145	0.032	0.551	0.099	0.97

where $E_{co,t}$ is employment, corrected for the 2021 break in the survey series, Σ_o and Γ_o the substitution and complementarity contents of one domain, built from the patents filed in the years of the period, averaged over the four-digit codes of the group and divided by their standard deviation across cells, μ_{ISCO1} fixed effects for the ISCO major group and δ_c country fixed effects. The digital and the green pair are estimated in separate regressions, weighted by base-year employment, with standard errors clustered by occupation group, on the cells observed in all three periods. Under the displacement reading of the task framework $\beta_\Sigma < 0$, and under the augmentation reading $\beta_\Gamma > 0$. Occupation series at this detail begin in 2011, with ISCO-08, so the full period is 2011–2023, which the table also splits into 2011–2019 and 2020–2023. Columns (1) to (3) use the LFS microdata cells at three digits for 21 countries, the 24 countries of the microdata extract less Bulgaria, Slovenia and Malta, which report occupation at one or two digits. Germany, Austria and the Netherlands are absent from the extract. Columns (4) to (6) use the public Eurostat table of employment by two-digit occupation for the same 21 countries, so that the test can be reproduced without access to the microdata.

Over 2011–2023 in the microdata, one standard deviation more digital complementarity content is associated with employment growth about 20 log points higher and one standard deviation more digital substitution content with growth about 12 log points lower, both significant at the 1 percent level. Green substitution content is negative and not significant (Table 7). Over 2011–2023 in the public table only the green substitution content is significant. Of the two sub-periods, the association appears in 2020–2023. Over 2020–2023 both digital contents carry their expected signs and are significant in both sources, with coefficients about half as large in the public table, and the

green substitution content is negative in both and significant in the public table. Over 2011–2019 no substitution content is significant, and the complementarity contents are significant only at 10 percent, in one source each. Extending the public table to all member states lowers the digital estimates for 2020–2023 to −0.035 and 0.027, both still significant. Without Germany, whose occupation series carries a break flag in the Eurostat table, they are −0.043 and 0.035, close to those on the 21 countries. The two sub-periods omit the change from 2019 to 2020, so columns (1) and (2) do not add up to column (3). Exposure alone is not associated with employment change in any period or source, and when the domain exposure is entered next to the two contents, substitution keeps its sign and significance over 2020–2023 in both sources and complementarity in the microdata, while over 2011–2023 in the microdata the three regressors, correlated at 0.7 to 0.9, separate less cleanly (Appendix D.9). Coding the second grade of the scale as a quarter of the reached content instead of none makes the 2020–2023 associations larger.

The respondent model was trained on text that postdates the outcome periods and may describe the occupational changes this test measures, so the test is not out of sample for the rater. If the ratings reflected knowledge of past employment trends, the association would appear in both periods, but over 2011–2019 no substitution content is significant and the complementarity contents only at 10 percent. This does not establish that the rater was uninformed by the outcomes.

Table 7: Substitution and complementarity contents and employment change across country-by-occupation cells. Employment change is over 2011–2019, 2020–2023 and 2011–2023, in the LFS microdata at three-digit ISCO in columns (1) to (3) and in the public Eurostat table at two-digit ISCO in columns (4) to (6), and the contents are built from the patents filed in the years of each period.

	Microdata, ISCO three-digit			Public table, ISCO two-digit		
Change in log employment	2011–2019	2020–2023	2011–2023	2011–2019	2020–2023	2011–2023
	(1)	(2)	(3)	(4)	(5)	(6)
A. Digital contents						
Digital substitution content	0.009	-0.081***	-0.125***	0.055	-0.044**	-0.023
	(0.027)	(0.017)	(0.046)	(0.051)	(0.021)	(0.059)
Digital complementarity content	0.055*	0.067***	0.196***	0.010	0.033**	0.064
	(0.033)	(0.013)	(0.048)	(0.036)	(0.015)	(0.043)
B. Green contents						
Green substitution content	0.022	-0.046	-0.088	-0.038	-0.047**	-0.088**
	(0.031)	(0.029)	(0.066)	(0.028)	(0.020)	(0.037)
Green complementarity content	0.003	0.007	0.067	0.061*	-0.001	0.068
	(0.039)	(0.036)	(0.068)	(0.034)	(0.029)	(0.045)
ISCO major-group fixed effects	✓	✓	✓	✓	✓	✓
Country fixed effects	✓	✓	✓	✓	✓	✓
Log base-year employment	✓	✓	✓	✓	✓	✓
Base-year employment weights	✓	✓	✓	✓	✓	✓
Occupation clusters	120	120	120	40	40	40
Observations	1,739	1,739	1,739	786	786	786

Change in log employment of country-by-occupation cells observed in all three periods, on the contents built from the patents filed in the years of the same period, averaged over the four-digit codes of the occupation group and divided by their standard deviation. The digital and the green pair are estimated in separate regressions. Columns (1) to (3) use the EU Labour Force Survey microdata and columns (4) to (6) the public Eurostat table of employment by two-digit occupation, for the same 21 EU countries. Occupation series begin in 2011, with ISCO-08. Standard errors clustered by occupation group are in parentheses. * $p<0.10$, ** $p<0.05$, *** $p<0.01$.

7 Conclusion

The direction a language model assigns to digital technologies, substitution or complementarity, is associated with employment change across European occupations. For green technologies, only substitution content in the public Eurostat table is significant. A language model replaces expert judgement at three points, in page selection and placement in the taxonomy, in review of the patent links behind exposure, and in the rating of direction. A technology hierarchy can be built from Wikipedia and patents with recorded manual input and measured coverage, and validated with classification codes and citations that the construction never uses. Its correction ledger and the record of its language-model review document how much structural judgement remained necessary. The exposure measures retain every technology node, and every correction by the LLM as judge is recorded. The direction measure is stable across draws, is set mainly by the technology and the occupation, and for digital technologies is associated with employment change across occupations in the LFS over 2011–2023.

The respondent model matches the experts' rating distributions but not their individual ratings, and it matches least well on substitution (Rony and Ciarli, 2026). Its training text may describe the outcomes it is tested against. The two statements are not matched in strength, replacing human workers against working alongside and supporting them, and both are phrased in the future tense while the employment test measures past change, so part of the gap between the two constructs may follow from wording. The industry enters the prompt only as the title of its NACE section, a coarse description of the context in which a technology meets an occupation. The system prompt casts the model as an expert in the domain's technologies before the persona is assigned, which may limit how far the personas move the ratings. Patents are weighted by their citations relative to their filing-year cohort, so the contents of the most recent period rest on the cited minority of its cohorts (Appendix C.6). Twelve green nodes carry descriptions the model cannot read as technologies (Appendix D.6). The task text comes from the English version of the European ESCO classification, and patents measure the supply of technology without recording its adoption. On the measurement side, the taxonomy's placement corrections and the adopted review proposals are curated and recorded, and the occupation side of the exposure measure carries a residual bias towards concretely worded tasks.

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A Data

The digital taxonomy is built from the English Wikipedia category graph, seeded with 27 curated technology categories, 25 of which yield pages. For every page it reaches, the crawl records the page title and the category path from the seed. The short description from the Wikipedia summary interface and the internal wiki-links of the article body are retrieved later, for the pages that survive the first cleaning stages. The links define the edges of the clustering graph, and the descriptions provide the edge weights and the text used in patent matching.

Three patent corpora are used, all drawn from one extract of patent applications with their DOCDB family (the European Patent Office's grouping of applications that share the same priorities), filing year, filing office, forward citations, CPC codes, title and abstract, in which each application is classified as digital or green. The taxonomy-matching and coverage corpus comprises about 1.9 million digital patents with PatentSBERTa abstract embeddings and is used for the page filtering and the coverage validation. The exposure corpora exclude applications filed only at the Chinese, Taiwanese or Russian office whose CPC codes no other office assigned. The digital exposure

corpus comprises some 880,000 patents with abstracts, titles, filing years, forward-citation counts and CPC codes. Patent tagging and the digital exposure measure use it. The green exposure corpus comprises about 5.7 million patents carrying OECD ENV-TECH indicators derived from their CPC codes.

Occupations are 427 of the 436 four-digit unit groups of ISCO-08, as implemented in the ESCO classification, which carries no occupations for the three armed-forces groups and for six residual groups not elsewhere classified. Each occupation is represented by its title and by task sentences parsed from the ESCO group descriptions, some 2,800 sentences in total, between six and seven per occupation, each combining the occupation title with one task. Industries are the 272 three-digit groups of NACE Rev 2, represented by some 850 activity sentences from the Eurostat NACE Rev 2 structure and its explanatory notes. The same occupation- and industry-side text underlies the digital and the green exposure measures, so that cross-domain comparisons hold the labour-market side fixed.

The grid of the substitution-complementarity measure is restricted to occupation-industry pairs that account for at least 0.1 percent of the occupation's pooled 2011–2023 employment in the LFS, at the level of three-digit ISCO groups by NACE sections. The grid spans 124 occupation groups, 20 sections and some 900 occupation-industry pairs, about 4,000 digital and 8,000 green triplets. The validation of the direction measure uses the change in employment of country-by-occupation cells over 2011–2019, 2020–2023 and 2011–2023, from the same microdata at three-digit ISCO groups and from the public Eurostat table of employment by two-digit occupation. Only cell-level aggregates enter this paper, and the microdata do not leave the restricted-access environment in which they are held. This paper is based on data from Eurostat, EU Labour Force Survey, 2011–2023. The responsibility for all conclusions drawn from the data lies entirely with the authors.

Two sentence-embedding models are used in fixed roles that Section 4.1 motivates. PatentSBERTa (`AI-Growth-Lab/PatentSBERTa`) (Bekamiri et al., 2024), a patent-domain sentence-BERT model (Reimers and Gurevych, 2019), embeds pages and patents for taxonomy matching and patent-to-technology membership, and the task and activity sentences for the page gate. The general-purpose sentence-embedding model all-mpnet-base-v2 (Song et al., 2020; Reimers and Gurevych, 2019), checkpoint `sentence-transformers/all-mpnet-base-v2`, provides the task-to-patent similarities that carry the value of the exposure measure. All embeddings are normalised to unit length, so that inner products equal cosine similarities, and large-scale similarity search uses the FAISS library (Johnson, Douze, and Jégou, 2021).

Table 8 records the model and settings of four steps at which a language model enters the construction. The labelling and the page corrections of the correction ledger and the first round of the taxonomy review also use a language model (Appendix B.7).

Table 8: Language-model steps of the construction.

Step	Model and access	Settings and record
Page judge (Appendix B.1)	Gemini 3 Flash, preview release (Google), through OpenRouter, early 2026	Temperature 0, top-p 1, pages in batches; prompt in the taxonomy pipeline code
Second review round of the taxonomy (Appendix B.7)	Claude Opus 5.5 (Anthropic), through Claude Code, October 2026	Proposals adopted by the authors after inspection; every change recorded in the taxonomy build
Link review (Appendix C.2)	Claude Opus 5.5 (Anthropic), through Claude Code, October 2026	Verdicts on named patents and links, recorded with the build
Direction ratings (Section 5.1)	DeepSeek-V3.1, open weights, through OpenRouter, September and October 2026	Temperature 0.3, top-p 0.3, five draws, JSON schema; prompt in Appendix D.1

B Construction of the technology taxonomies

B.1 Crawl, cleaning, matching and filtering for the digital taxonomy

The digital taxonomy is constructed in four data stages, followed by clustering and validation. Candidates are generated broadly and filtered by a cascade of increasingly expensive filters. Table 1 in Section 2.1 reports the reduction funnel, from 1.8 million crawl records to about 8,000 pages. The cheap early filters remove obvious noise in bulk, and the patent-based and model-based filters at the end decide the ambiguous cases.

A breadth-first search over the category graph proceeds from the seeds to a depth of four category levels, queuing each subcategory for a visit and recording the article pages. A second, depth-first pass writes one record per seed, category path and page, which keeps the path by which each page was reached for the later filters on category paths. The 1.8 million records hold some 300,000 distinct pages, reached on average through six category routes, and include biographies, films and geographic entries that share categories with technology topics.

A five-stage cascade, ordered by computational cost, reduces the crawl output by 96 percent to about 78,000 pages. The cascade first filters titles and category paths on structural patterns (disambiguation suffixes, entertainment, biographical and geographic patterns, and path-level exclusion of company, biography and conference branches). It then retrieves the descriptions of the surviving pages and computes a keyword-based relevance score, which removes no page. Filters on the descriptions next target biographies, organisations, media and events, and a last stage extracts the body wiki-links of each remaining page, which are stored as the edges of the subsequent clustering graph. Duplicate titles are resolved by retaining the first occurrence in crawl order.

The descriptions of the cleaned pages that pass a first noise filter, some 71,000, and the 1.9 million patent abstracts are embedded with PatentSBERTa. An approximate nearest-neighbour

index, of the inverted-file product-quantisation type with 4,096 lists and 64 subquantisers of 8 bits, retrieves 100 candidate patents per page, and exact cosine similarities are then recomputed on the original vectors, so that retrieval is approximate but all reported similarities are exact.

Three filters reduce the cleaned set to the candidate pages. The noise filter on titles applies some 120 title patterns and a curated list of over 2,600 excluded titles accumulated through manual review. A patent-similarity threshold retains pages whose best-matching patent reaches a cosine similarity of 0.6, removing pages whose wording is close to technology topics but which are unrelated to patented technology. Some 12,000 pages pass both filters. A language model, Gemini 3 Flash (Table 8), then classifies whether each remaining page describes an actual technology and accepts 8,069 pages. A curated list of noise titles, compiled by inspecting preliminary clusterings, removes 58 further pages before clustering.

B.2 Coverage diagnostics

The coverage diagnostics are computed for two representations of the patents, embeddings of their titles and of their abstracts, each matched against the page descriptions, and Table 9 reports both. The abstracts raise the mean top-1 similarity, the similarity of a patent to its closest page, from 0.517 to 0.565 and coverage at the 0.5 threshold from 59 to 94 percent. The CPC and citation criteria change little between the two, so the matching signal does not depend on the patent text field. The higher citation-modularity ratio of the titles, the within-page relative to the between-page citation rate, is computed on a much smaller set of covered patents. The connected components of the uncovered patents separate the two. With titles almost half of the uncovered patents form one citation-connected component, a systematic blind spot, whereas with abstracts the largest component holds some 3,200 patents. Figures 11 through 14 display the threshold sensitivity, the external validity diagnostics, and the component-size distribution.

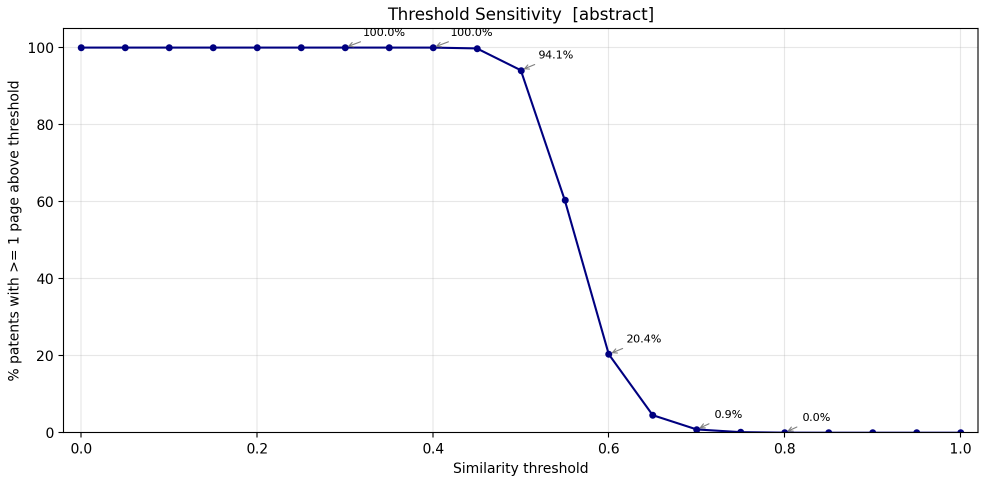


Figure 11: Patent coverage as a function of the similarity threshold (patent abstracts). Lowering the threshold from 0.5 to 0.45 adds 5.7 percentage points of coverage, while raising it to 0.55 removes 33.7.

Table 9: Coverage diagnostics with patent titles and with patent abstracts matched against the page descriptions. The sample is 1,911,137 patents and 8,069 pages, and the threshold is 0.5. Citation edges count each citation in both directions.

Statistic	Patent titles	Patent abstracts
Mean top-1 similarity	0.5167	0.5651
Median top-1 similarity	0.5124	0.5611
Standard deviation of top-1 similarity	0.0587	0.0463
Coverage at threshold 0.5 (%)	58.76	94.11
Coverage at threshold 0.6 (%)	8.60	20.39
Pages closest to a covered patent with CPC codes	7,601	7,701
Patents below threshold	788,181	112,485
Within-page CPC Jaccard similarity	0.2832	0.2828
Between-page CPC Jaccard similarity	0.1015	0.1043
CPC coherence ratio	2.79	2.71
Above-threshold patents (citation analysis)	1,122,956	1,798,652
Citation edges	4,563,026	10,048,412
Within-page citation edges	463,936	740,920
Citation-modularity ratio	61.53	56.55
Below-threshold components of size ≥ 10	44	111
Patents in such components	379,160	6,966
Largest component	378,607	3,245

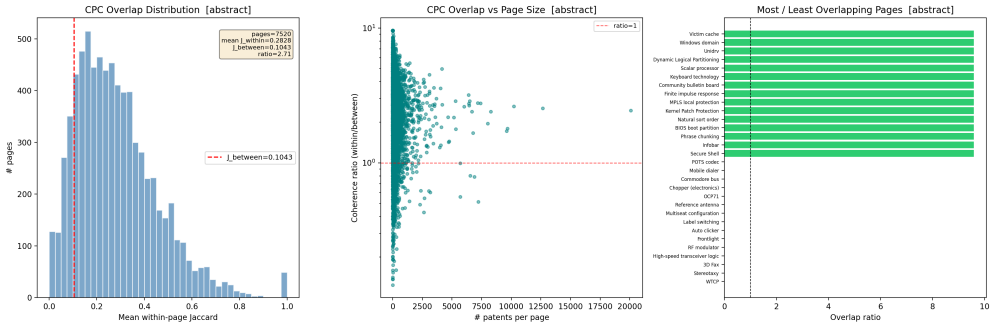


Figure 12: Within-page against between-page CPC code overlap (Jaccard similarity). Pairs of patents matched to the same page have three-digit CPC sets 2.71 times as similar as pairs matched to different pages. The CPC classification is assigned independently of the embeddings.

B.3 Hierarchical community detection

The filtered pages, 8,011 after the removal of the listed noise titles, form the graph of Section 2.2. In the configuration search each edge carries the weight

$$w_{uv} \;=\; sw \cdot \max\{0,\ \cos(e_u, e_v)\} \;+\; (1 - sw) \cdot \frac{n_{uv}}{\max n}, \tag{7}$$

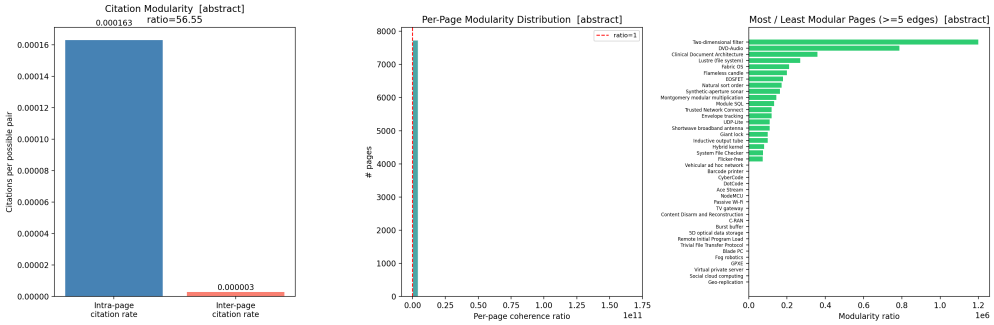


Figure 13: Citation modularity (patent abstracts). The within-page citation rate per pair of patents, 1.63×10^{-4}, exceeds the between-page rate, 2.88×10^{-6}, by a factor of 56.5.

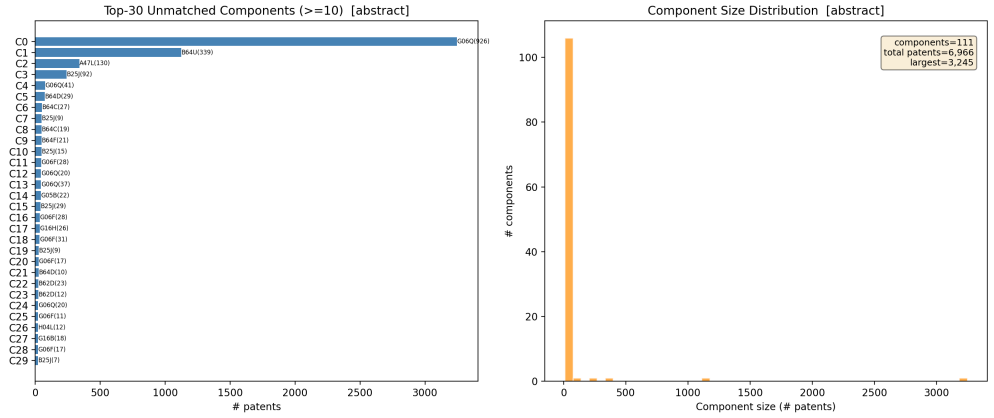


Figure 14: Size distribution of citation-connected components among below-threshold patents (patent abstracts). The largest component of the 112,485 uncovered patents holds 3,245.

where n_{uv} is the number of wiki-links between the two pages, normalised by its maximum over the graph, and sw is the semantic weight. An optional degree penalty divides each weight by $\log(1 + \deg u)\log(1 + \deg v)$. The selected configuration sets $sw = 1$, so that the link graph contributes the set of candidate edges and all weights are cosine similarities, prunes edges below $\tau_{\text{edge}} = 0.20$, uses no degree penalty and a margin weight $\lambda_m = 8$ in Equation (3), and applies the single resplitting round of Section 2.2. Leiden runs with the resolution-weighted configuration-model partition of Equation (2), thirty iterations and seed 42. Within each resolution range the binary searches take thirty steps between $\gamma = 10^{-4}$ and $\gamma = 10$, and the within-cluster similarity of the score is computed on up to 200 pages per child.

Before the resplitting round, the run attains positive within-minus-between margins at all three levels, +0.070, +0.027 and +0.040, coherence close to 0.70 at every level, and adjusted Rand indices over three seeds of 0.69, 0.88 and 0.84. After it, the raw output comprises 12 root, 56 intermediate and 205 base clusters, besides a peripheral group under each parent, on 7,479 core and 532 peripheral pages.

B.4 Configuration search

The clustering configuration was selected through grid searches over the semantic weight sw of Equation (7), the edge-pruning threshold τ_{edge}, and the margin weight λ_m of Equation (3), followed by a comparison of seven shortlisted configurations with and without one resplitting round and of a second resplitting round for three of them. The tables name each configuration by these three values, written sw, tau and mw.

A positive weight on link counts produces incoherent agglomerates. At $sw = 0.8$, the 20 percent weight on the normalised link count allows the dense cross-linking of computing pages to connect semantically unrelated topics through short link chains. The result is a single incoherent cluster of 1,814 pages at the intermediate level and 1,641 at the base level, spanning memory hardware, printer drivers, checksum algorithms, power supplies and debugging tools, together with a 2,428-page root cluster mixing optics, radio-frequency technology, imaging and robotics. At $sw = 1.0$ the largest clusters are thematically homogeneous and were subdivided.

Table 10 reports the final shortlist by the bottleneck criterion, the worst-level within-minus-between margin. The $sw = 0.8$ configurations attain higher average margins but lower worst-level margins, and carry the incoherent agglomerate described above. Table 11 reports the internal and external evaluation of three shortlisted configurations with one and with two resplitting rounds. The second round changes the criteria of the selected configuration little, and the single round was retained. Table 12 reports the size distributions. Within the retained family, the margin weight is immaterial, $\lambda_m = 8$ and $\lambda_m = 10$ differing by one or two base clusters at identical worst-level margins, and $\lambda_m = 8$ was adopted. The experiment log displays the search graphically.

Table 10: Final grid-search shortlist, ranked by the worst-level within-minus-between margin, before the resplitting round. The intermediate level binds for the $sw = 1.0$ configurations.

Configuration	sw	τ_{edge}	λ_m	Worst margin	Average margin
sw1.0 tau0.2 mw8 (selected)	1.0	0.2	8	+0.0266	+0.0453
sw1.0 tau0.2 mw10	1.0	0.2	10	+0.0266	+0.0454
sw0.8 tau0.2 mw6	0.8	0.2	6	+0.0227	+0.0488
sw0.8 tau0.3 mw8	0.8	0.3	8	+0.0224	+0.0467
sw0.8 tau0.2 mw4	0.8	0.2	4	+0.0211	+0.0483

Table 11: Evaluation of shortlisted configurations at the base level, with one (single) and two (double) resplitting rounds. Base K counts the peripheral group under each parent as a cluster. Coh denotes the median semantic coherence, and PD the share of pages closer to their own than to the nearest other centroid and the mean page-level margin. CPC4 and Cite denote the medians over clusters of the four-digit CPC ratio and of the citation ratio, and Expert the expert-agreement percentage.

Configuration	State	Base K	Coh	PD (%)	PD mean	CPC4	Cite	Expert
sw0.8 tau0.3 mw8	single	103	0.744	61.2	+0.0187	1.635	76.5	45.3
sw0.8 tau0.3 mw8	double	138	0.790	61.8	+0.0224	1.632	79.7	46.4
sw0.8 tau0.2 mw4	single	149	0.733	60.9	+0.0235	1.689	65.7	52.3
sw0.8 tau0.2 mw4	double	170	0.733	61.6	+0.0254	1.665	59.1	55.0
sw1.0 tau0.2 mw8	single	215	0.764	64.1	+0.0282	1.650	75.2	59.3
sw1.0 tau0.2 mw8	double	240	0.765	64.7	+0.0307	1.650	80.2	59.9

Table 12: Cluster-size distributions of the two leading configuration families after one resplitting round. Gini denotes the Gini coefficient of cluster sizes, and Med/Max the ratio of the median to the maximum cluster size. The selected family has more balanced root clusters.

Configuration	Level	K	Gini	Med/Max
sw0.8 tau0.3 mw8	Root	9	0.593	0.049
sw0.8 tau0.3 mw8	Intermediate	32	0.725	0.012
sw0.8 tau0.3 mw8	Base	94	0.812	0.006
sw1.0 tau0.2 mw8	Root	12	0.537	0.338
sw1.0 tau0.2 mw8	Intermediate	56	0.691	0.028
sw1.0 tau0.2 mw8	Base	205	0.774	0.008

B.5 Direct clustering of patent embeddings

The principal alternative to page-based tagging is to cluster the patent abstract embeddings of the exposure corpus directly by spherical k-means (Dhillon and Modha, 2001; Lloyd, 1982), which assigns every patent by construction. On the full corpus, over k between 10 and 1,000, the CPC ratio and the citation ratio rise with k throughout, and the silhouette index (Rousseeuw, 1987), computed on a sample of 25,000 patents, peaks at $k = 20$ and declines thereafter, so that neither criterion identifies a number of technologies. The direct clusters are unlabelled partitions estimated from the corpus alone, and they omit externally documented technologies that do not form separable patent clusters. They therefore serve as a benchmark for the taxonomy's tagging and cannot replace its named structure. The experiment log records the sweeps, including the effects of removing the common direction of the embedding space and of reducing its dimension before clustering.

B.6 Validation of the selected clustering

Tables 13 through 16 report the four validation panels of the selected run. The margin of a cluster (Table 13) is the mean similarity of its pages to its centre less the similarity of its centre to the nearest other centre of the same level. Margins are negative for two thirds of the root and almost three quarters of the intermediate clusters and for half of the base clusters, and they fall with cluster size (Figures 16 and 17). The negative margins belong to large clusters with low within-cluster similarity, the positive margins to small specialised clusters. Tables 14 and 15 report CPC homogeneity and citation coherence. The latter declines mechanically with depth as clusters shrink and did not enter model selection. Restricting the coverage audit to the pages in non-peripheral clusters (Table 16) reduces coverage by about one percentage point.

Table 13: Cluster margins of the selected run. The margin of a cluster is the mean similarity of its pages to its centre less the similarity of its centre to the nearest other centre of the same level. The peripheral group under each parent counts as a cluster.

Level	Clusters	Share negative	Mean margin	Median margin
Root	12	66.7%	−0.085	−0.233
Intermediate	58	72.4%	−0.087	−0.154
Base	215	52.1%	−0.018	−0.014

Table 14: CPC homogeneity of the selected run. The modal-CPC share is the share of the most frequent three-digit CPC code in a cluster's CPC assignments, and HHI is the Herfindahl–Hirschman index of concentration over CPC codes.

Level	Mean modal-CPC share	Mean HHI
Root	0.443	0.270
Intermediate	0.446	0.285
Base	0.438	0.278

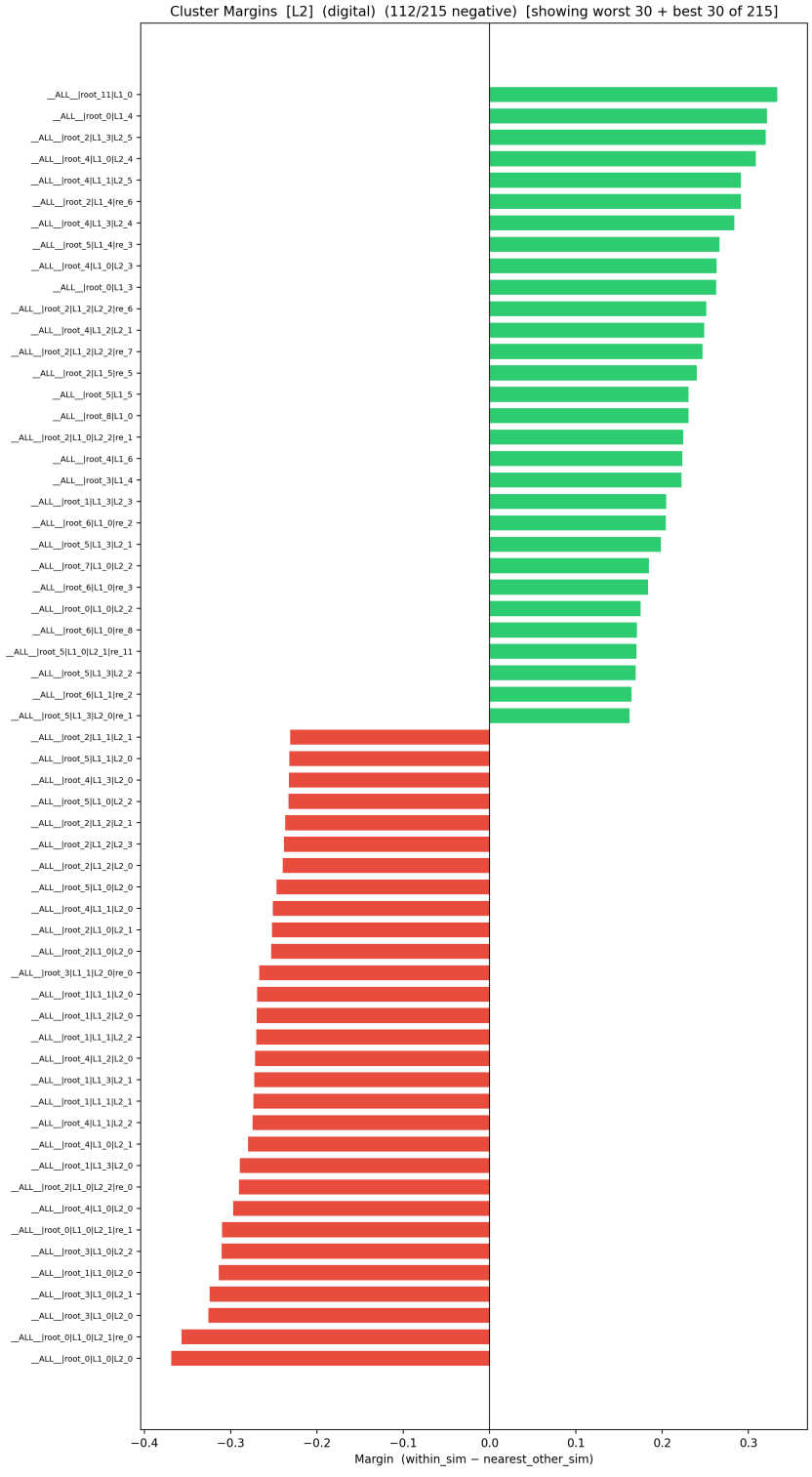


Figure 15: Margins of the 30 base-level clusters with the lowest and the 30 with the highest margins, of 215 in all. Figure 16 relates the margins to cluster size.

Table 15: Citation coherence of the selected run. The measure is the mean share of a cluster's citation edges that remain within the cluster.

Level	Mean within-cluster citation rate
Root	0.272
Intermediate	0.136
Base	0.085

Table 16: Coverage before clustering (all pages) and on the pages in non-peripheral clusters of the selected run.

Statistic	All 8,069 pages	7,435 clustered pages
Mean top-1 similarity	0.565	0.563
Coverage at threshold 0.5 (%)	94.1	93.1
Citation-modularity ratio	56.5	52.7

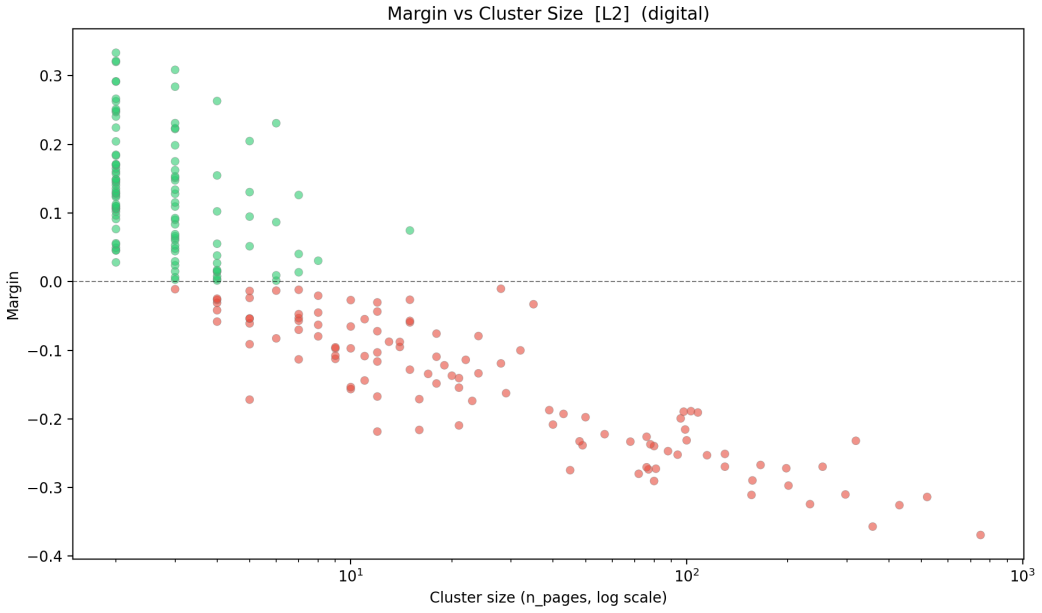


Figure 16: Margin against cluster size at the base level. Margins fall with cluster size.

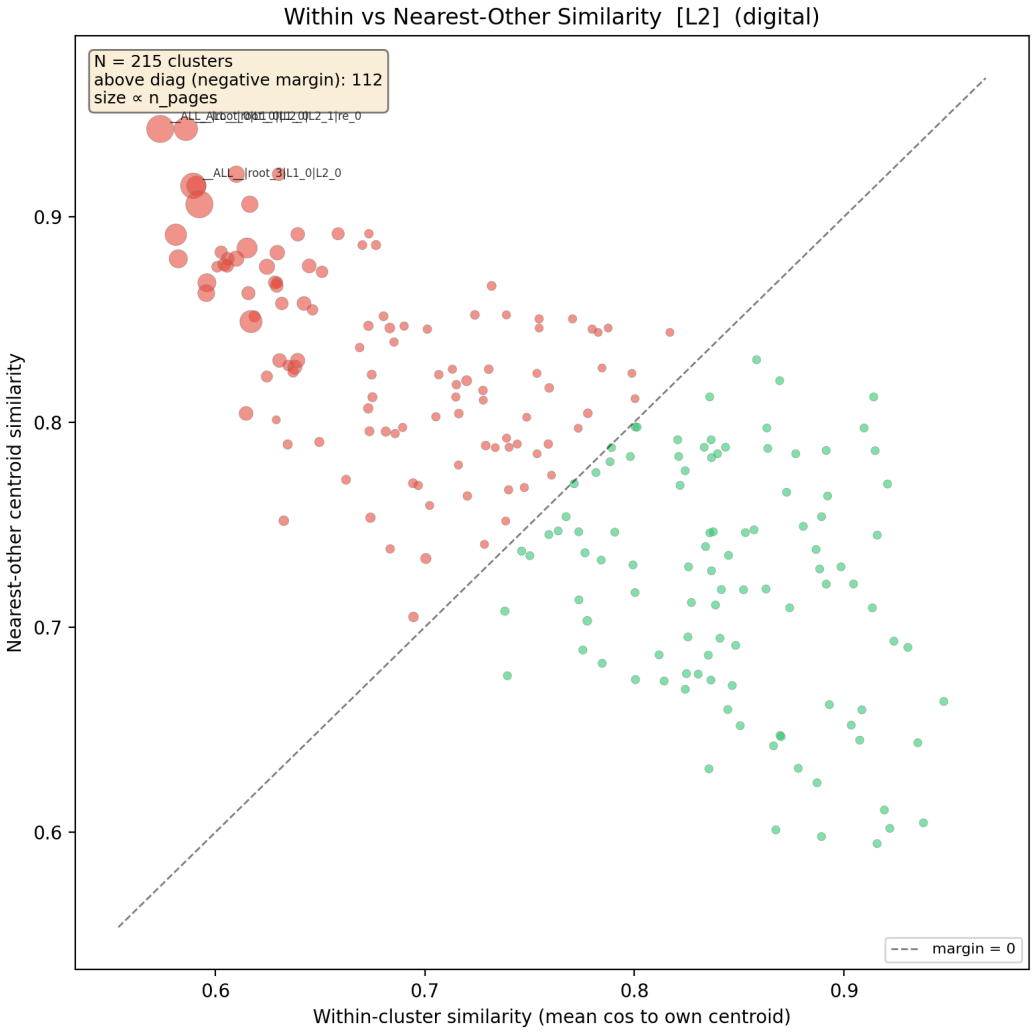


Figure 17: Within-cluster similarity against the similarity of the cluster's centre to the nearest other centre at the base level. Point size is proportional to the number of pages. Clusters above the diagonal have a negative margin. The large clusters lie above the diagonal, the small clusters below it.

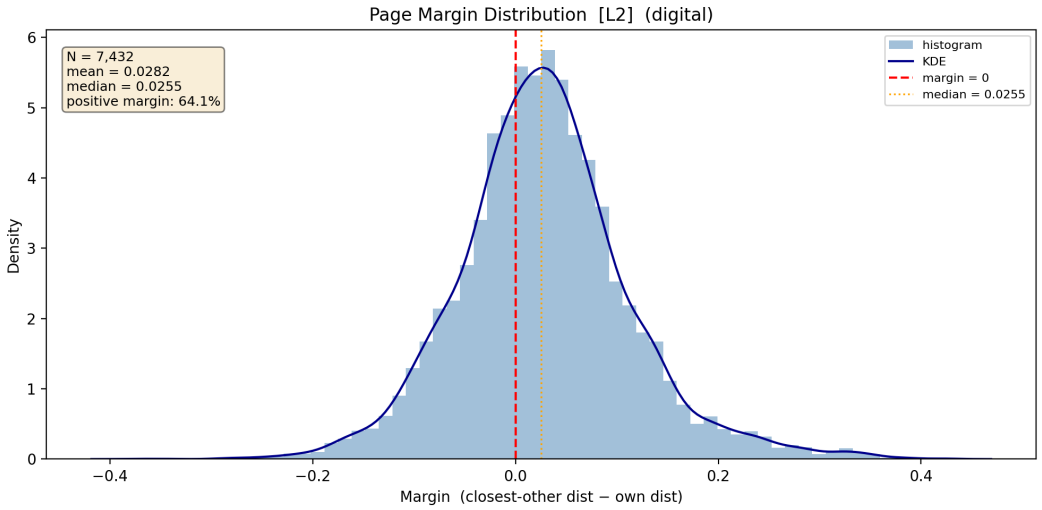


Figure 18: Distribution of page-level margins at the base level.

B.7 From clustering to the final taxonomy

The final taxonomy is obtained from the raw clustering in two stages, a documented correction ledger and a language-model review of the pages of every technology. The correction ledger comprises bottom-up labelling by a language model, a first correction pass and a second pass. The first pass combines automated consistency repairs, twelve blocks of curated structural corrections and page-level corrections, the latter two amounting to 1,839 logged corrections. The second pass splits 30 base clusters into 62, applies 1,130 verified page reassignments and 8 removals, and places 97 orphaned pages by language-model adjudication, leaving 7,398 pages. Appendix B.8 reports the ledger. Four of the ten root domains originate in this stage, through two splits of raw roots by content. Across both passes the ledger removes 613 of the 8,011 clustered pages.

The language-model review then reads the structure technology by technology and judges, on technology content alone, whether each page belongs to its technology, to another technology or to none. Its proposals were adopted by the authors after inspection, in two rounds. The first review round reads the member pages of every technology and flags 95 pages in 38 technologies as off-topic. Of these, 23 belong to the same field and stay, 22 are moved to the technology they describe, 42 are removed, and 8 pages on domestic, laboratory and surgical robots, which the clustering had placed with swarm and legged robots, form two new technologies, one for service robots and one for medical and laboratory robots. The same round removes six pages of the Quantum Computing root that collect patents through word collisions, among them Cloud-based quantum computing, and restores the Cloud computing page, which the language-model filter of the page selection had removed as a computing paradigm. After the first review round 7,351 pages remain.

In the second review round Claude Opus 5.5 (Anthropic) reads every base technology with the titles of its pages and proposes corrections, each tied to named pages, for pages placed in the wrong technology, pages that are not digital technologies, duplicate pages, and names that misdescribe

the pages they cover. The proposed merges that would only have coarsened the granularity were not adopted. The second review round moves 169 pages to another technology, 66 individually and 103 in five splits. Of the latter, 99 form five new technologies, Avionics and Inertial Navigation, Building Automation and Smart Home, Smart Grid and Utility Automation, Wearable Devices, and Blockchain and Distributed Ledgers, and 4 join the existing satellite-navigation technology. One technology, EM Wave-Material Interaction, is dissolved. Two of its five pages join the imaging-sensor technology and the other three are among the pages removed below. The second review round removes 24 pages that are not digital technologies, such as a dating method, a pipe fitting for thermometers, the mimeograph and offset printing, and the 120 remaining pages of five technologies that are not digital, among them power electronics and electrical generators and dynamos. Each of 199 groups of pages with identical description embeddings, such as Smart band and Fitness tracker, is reduced to one page, which removes 279. One technology is merged into another, and eleven are renamed to describe the pages they hold. The round regroups the intermediate level and adds a tenth root, Transport and Navigation Systems, which separates 201 pages of vehicle electronics, driver assistance, traffic and railway management, satellite navigation and avionics from the Robotics and Manufacturing root. In all, the second review round changes the root of 500 pages and the intermediate cluster of 2,931, and leaves 6,928 pages. The patents of a removed page count towards no technology, except those of a removed duplicate, which pass to the page kept.

B.8 The correction ledger

The first correction pass consists of automated phases and a curated phase. The automated phases carry out the merges proposed by the language-model labeller, resolve peripheral pages, repair violations of the parent-child hierarchy and overlaps between sibling clusters, and normalise the structure, which leaves 164 base-level entries. The curated phase comprises twelve blocks, which merge six duplicate base clusters, absorb four single-child roots, repair two misplacements, relabel three clusters whose content contradicted their labels, split one root into an imaging-and-display and a robotics-and-manufacturing domain, relocate five misplaced fragments and one 203-page web-standards cluster, relocate a 321-page radar cluster from the artificial-intelligence to the signal-processing domain, relabel two irreducibly broad clusters, split a 440-page deep-learning-and-language cluster into its neural-network and language components, reorganise the artificial-intelligence branch into five method-based intermediate clusters, consolidate the computing-hardware branch, and split the networking root into a cryptography-and-security and a networking-and-telecommunications domain. Together with 1,745 page-level corrections proposed by a language model, 312 of them applied after a manual check and the rest, rated high-confidence, applied directly, the first pass records 1,839 corrections and leaves 153 base clusters on 7,406 pages.

The second pass refines the structure at page level. It splits 30 base clusters into 62 and applies 1,130 verified page moves, of which 686 are placed in newly created subclusters through a page-by-page assignment list and 444 by direct assignment. It executes 8 verified removals, does not apply 68 rejected suggestions and places 97 orphaned pages by language-model adjudication over

the complete taxonomy in batched calls. The ledger leaves 7,398 pages, with the page balance reconciled exactly. The language-model review of Appendix B.7 starts from this structure.

B.9 Patent tagging schemes

The assignment of the patent corpus to the taxonomy, as it stood after the correction ledger, was audited across six tagging schemes. The tagging step of the taxonomy construction assigns each patent the base technology of its best-matching page subject to a similarity floor of 0.50 and covers 94 percent of the exposure corpus. The exposure measure uses the membership rule of Section 4.1 instead. Judged by hand on the page matches in the band immediately above the floor, counting all qualifying matches and not only the best, precision is about 0.20, 6 of 30 sampled matches (Figure 19). The dominant error mechanism is polysemy, a patent matching a page on the wrong sense of shared vocabulary. Re-tagging by revealed comparative advantage (Balassa, 1965) rewards membership in rare clusters. Consensus rules, requiring either one strong match or a quorum of weak matches, raise precision but reduce coverage to between 44 and 80 percent. Tagging patents to cluster descriptions written by a language model attains a precision of 0.50 in the floor band at 53 percent coverage. Direct k-means assigns every patent but severs the connection to named technologies. The CPC classification cannot serve as the tagger. Its codes are nearly universal in the corpus, but the patents of a code spread over many base technologies, the leading one holding less than a fifth of them for most codes, and finer codes concentrate them little more (Figure 21). It can mark a patent as belonging to the domain but cannot assign it to a technology. No scheme attains high coverage and floor-band precision together. The exposure measure inherits this constraint through its membership relation, and the LLM as judge of Appendix C.2 addresses the entity-level consequences.

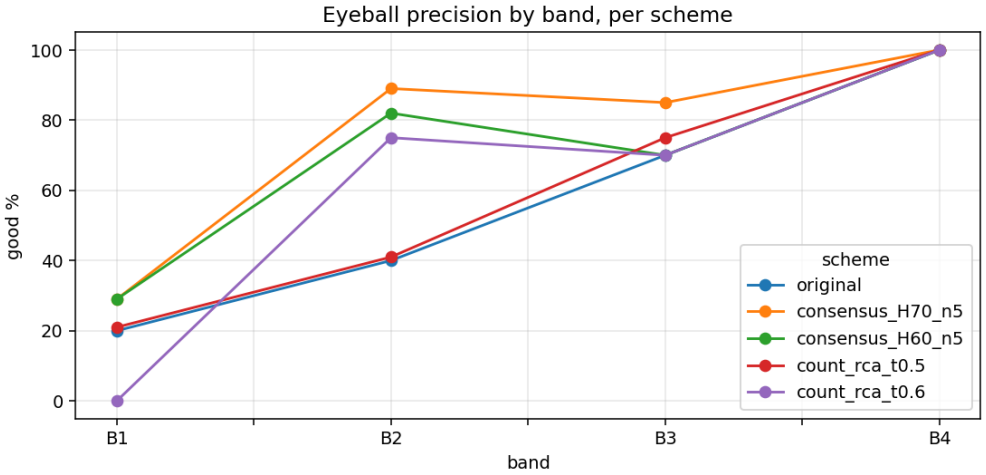


Figure 19: Tagging precision judged by hand, by similarity band. Precision is in percent, by band from the floor (B1) upwards, for page matching (original), two consensus rules and re-tagging by revealed comparative advantage at two thresholds. Page matching attains about 20 percent in the band above the 0.50 floor.

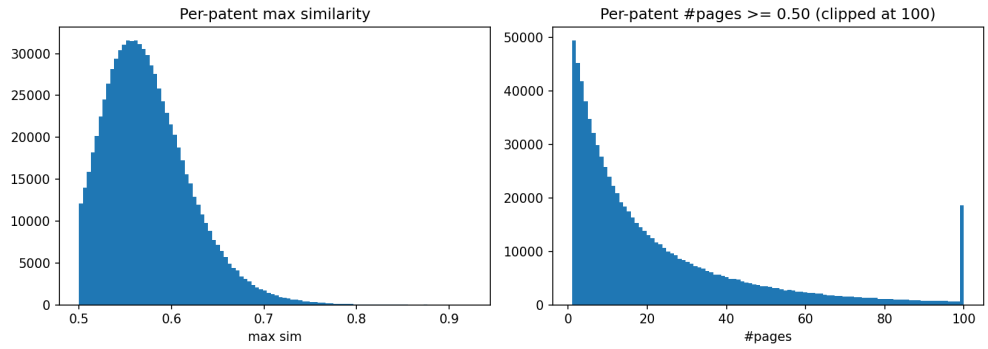


Figure 20: Distribution of qualifying page matches per patent at the similarity floor. Multiple membership at the floor is common, which compounds the polysemy mechanism.

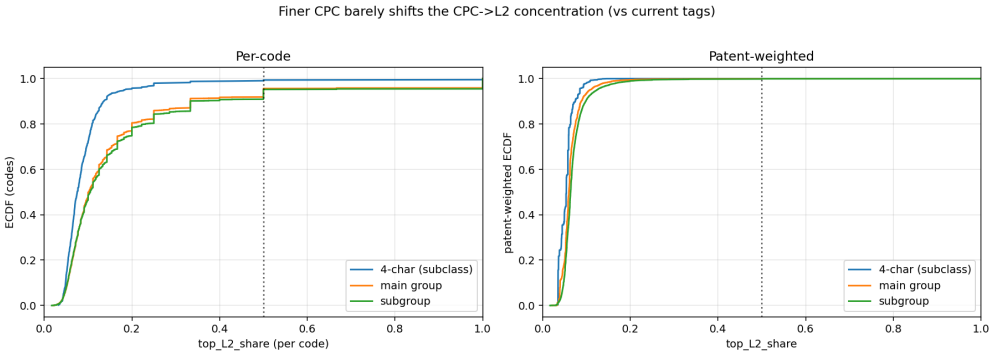


Figure 21: Concentration of CPC codes on base technologies, on the page matches. The panels show the distribution over codes of the share of a code's patents in its leading base technology, for four-character codes, main groups and subgroups, per code (left) and weighted by patents (right).

B.10 The OECD ENV-TECH green taxonomy

The green taxonomy is the OECD ENV-TECH classification (Haščič and Migotto, 2015), with 66 technology nodes defined by mappings from CPC codes and organised into ten groups. The groups are those for environmental management, for energy generation, transmission and distribution, for greenhouse-gas capture and storage, for transportation, for buildings, for wastewater and solid waste, for production, industry and agriculture, for ICT, for climate-change adaptation, and for smart grids, which carries its own Y04S codes and is treated as a separate group because the numbered subcategories of group nine describe climate-change adaptation and do not cover smart grids. A patent belongs to a node if and only if its CPC-derived ENV-TECH indicator is set. Multiple membership is permitted, and membership follows from the classification codes that patent examiners assign. The patent corpus encodes three parent nodes only through subordinate codes (water pollution control through three subcodes, waste management through five, and smart grids through five Y04S codes), and crediting these subcodes to their parent nodes recovers some 745,000 patents, 13 percent of the green corpus, that a literal reading of the parent indicators would omit.

C Construction of the exposure measures

C.1 The green measure

The green measure follows the digital measure except in membership. The member set $P_n(w)$ is given by the ENV-TECH indicators instead of page matching, so that no membership threshold and no page gate apply, and the value relation uses the same MPNet embeddings of the same occupation and industry sentences, with the floor at the 99th percentile of the green sentence-patent similarities, 0.32.

C.2 The LLM as judge

This appendix records the two steps of the LLM as judge of Section 4.2 and their effect.

In the digital domain the LLM as judge acts on patent links and on cells. First, it reviews the links that carry the largest shares of each technology's exposure, a link being one patent's contribution to one industry's or one occupation's exposure to the technology. For every technology exposed over 2009–2023, up to the fifty largest industry links and up to the fifty largest occupation links are read together with the technology's pages and each patent's title and abstract. The 15,337 links read in 181 technologies carry a median 44 percent of a technology's industry exposure and 46 percent of its occupation exposure in the measure on which the review was run. Each patent is kept in its technology, moved to the technology to which it belongs or removed when it is not a digital technology. Each link is kept when the industry or occupation plausibly produces, uses or is affected by the patent's technology and set to zero when the connection runs only through shared words or generic phrasing, as between television programmes and software programming or between warehouse storage and data storage. The verdicts keep 56 percent of the reviewed links and judge 49 percent of the reviewed patents to belong to their technology. They are attached to patents and carried over to the final taxonomy, where a verdict follows its patent to the patent's technology. A move into a renamed, merged or split technology goes to its successor, a move into a removed technology becomes a removal, a move into the technology the patent now occupies becomes a decision to keep it, and the verdicts on patents of removed pages lapse. On the final taxonomy the verdicts move 3,854 patents, remove 292 and set 6,434 links to zero. Together with the twelve cells per side that their moves newly expose to the cell list of the second step, they lower the sum of the untransformed digital exposure values, computed after the listed cells are set to zero, by 4.4 percent on the industry side and 4.3 percent on the occupation side. Instances of Claude Opus 5.5 (Anthropic), run through Claude Code in October 2026, produced the verdicts, in a first round on the five largest sentence-patent pairs of every technology, for occupations and for industries, and in a second round on the links. No person has audited a sample of the verdicts.

Second, it sets to zero the entity-technology cells that a reading of the complete rankings finds driven by a lexical coincidence or captured from another domain, such as refuse sorters ranked on a data-structures node through the word sorting, medical occupations on a calculator-software node, the forestry and logging industry on a system-logging node and the manufacture of wiring

devices on superconducting qubits through the word circuit. These are 133 occupation-side and 93 industry-side cells, 1.5 percent of the 15,470 exposed digital cells. The cell list identifies technologies by name. On the final taxonomy it follows renamed and merged technologies and drops the entries of removed ones.

In the green domain, membership is fixed by classification codes and the LLM as judge reduces to the second step. The reading of the green rankings for occupations and for industries identifies a single off-topic family ranked on the transport nodes through the word vehicle, the two occupations of drivers of animal-drawn and hand-propelled vehicles and the manufacture of transport equipment not elsewhere classified, hand-propelled and animal-drawn vehicles. Setting 18 occupation-side and 9 industry-side cells to zero, 0.07 percent of the 38,177 exposed green cells, resolves it, after which the electric and hybrid vehicle nodes are led by electrical engineers, locomotive drivers and motor-vehicle mechanics. Appendix C.7 describes the records of these corrections and gives examples.

No technology is dropped for lack of exposure. Technologies without any exposed entity, seven in the digital domain such as Quantum State Tomography and Optical Polarization Control, are kept with zero exposure. Every correction is recorded, the taxonomy review as verdicts on named pages, the link review as verdicts on named patents and links and the cell step as a list of entity-technology cells, so the measure can be rebuilt from the output of Equation (4) and these records. Only the cell step leaves the set of patents unchanged.

In the digital domain, the off-topic exposure that these corrections remove arises in the text-based attribution of patents to technologies, in which polysemous pages and abstract nodes attract unrelated patents.

C.3 Similarity scales of the three relations

The exposure measure rests on three similarity relations, page to patent, patent to task, and task to page. Table 17 reports the location of each relation's similarity distribution for the two embedding models that the measure uses. The page-patent relation is measured over the complete page-by-patent similarity matrix, $7{,}398 \times 876{,}009$ pairs, and the patent-occupation and occupation-page relations as each occupation's maximum over its task sentences, the former on a sample of about four million patent-occupation pairs. The two models operate on different scales, with PatentSBERTa similarities approximately twice MPNet's. Within each model the three relations differ, with the occupation-page relation lowest. The conventional threshold of 0.50 exceeds the 95th percentile of every relation under both models. As an absolute cut it selects extreme tails, deeper for MPNet than for PatentSBERTa and of different depths across the three relations. Figure 22 displays the distributions.

The same scale problem governs the choice of the embedding model for the value relation. Table 18 reports, for the three candidate embedding models, the share of pairs of a task sentence and a patent that survive an absolute threshold, on a random sample of five million pairs. PatentSBERTa and MPNet keep a fifth or less of the pairs at 0.3 and almost none at 0.5, whereas SPECTER2, an embedding model for scientific documents that was loaded without its task-specific adapter, keeps

Table 17: Location of the similarity distributions of the three relations, by embedding model. The columns report the mean and upper percentiles of the cosine similarity.

Model	Relation	Mean	p50	p60	p70	p80	p90	p95
PatentSBERTa	Page–patent	0.294	0.287	0.304	0.323	0.346	0.377	0.404
PatentSBERTa	Patent–occupation	0.288	0.286	0.302	0.319	0.340	0.369	0.395
PatentSBERTa	Occupation–page	0.247	0.244	0.259	0.276	0.297	0.327	0.354
MPNet	Page–patent	0.137	0.124	0.144	0.167	0.195	0.238	0.278
MPNet	Patent–occupation	0.158	0.154	0.173	0.194	0.220	0.260	0.297
MPNet	Occupation–page	0.125	0.119	0.136	0.155	0.178	0.215	0.250

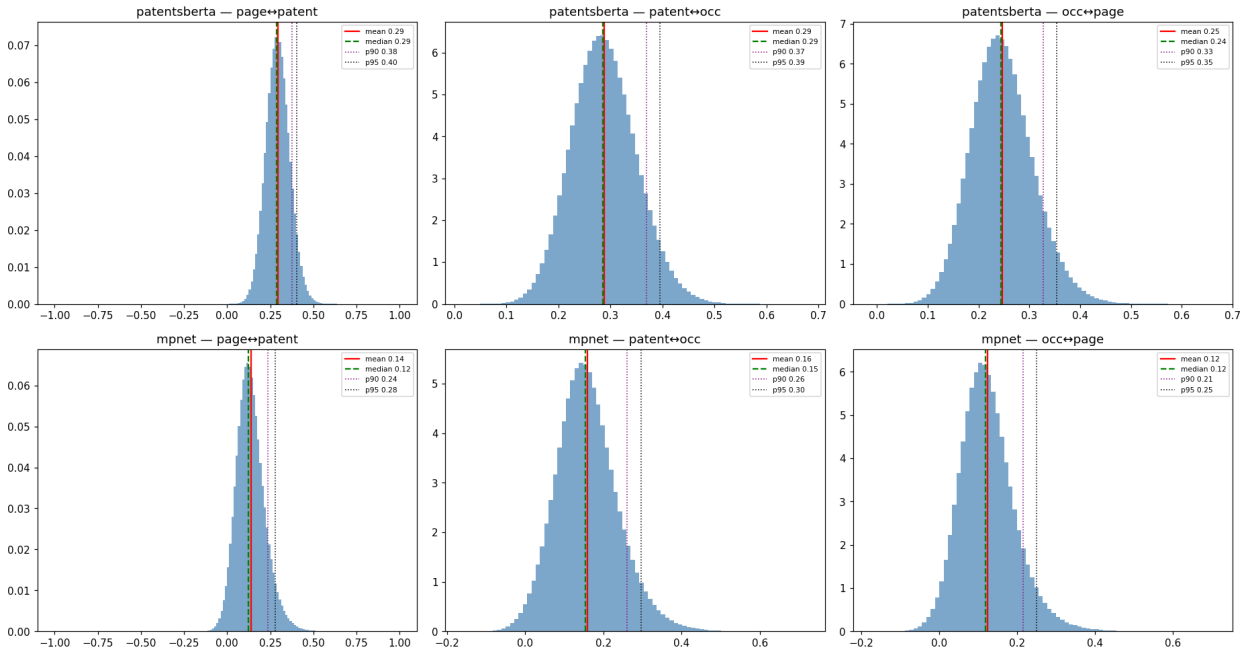


Figure 22: Similarity distributions of the page-patent, patent-occupation and occupation-page relations. The upper row shows PatentSBERTa and the lower row MPNet, with means, medians and upper percentiles marked. An absolute threshold of 0.50 lies beyond the 95th percentile of every relation under both models.

almost every pair at every threshold. Its similarities are compressed into a narrow high band in which neither a threshold nor a rank filter is informative. This configuration was excluded, and SPECTER2 with its adapter was not evaluated.

Table 18: Share of (task sentence, patent) pairs surviving an absolute similarity threshold θ, by embedding model. The sample is five million random pairs.

θ	PatentSBERTa	MPNet	SPECTER2
0.3	19.76%	2.31%	100.00%
0.4	1.76%	0.28%	100.00%
0.5	0.05%	0.03%	99.998%

Sweeping a shared absolute threshold across the three relations degrades the diagnostics of the measure jointly as the cut deepens, whereas per-relation percentile cuts leave them stable for both models. The experiment log records the sweeps. Every threshold is therefore defined on the percentile scale of the relation and model to which it applies, and the value floor $t_a = 0.32$ is the 99th percentile of the MPNet task-patent distribution.

C.4 The aggregation over sentences

The measure of Equation (4) averages over an entity's sentences the citation-weighted sum over a technology's member patents. Within a sentence the sum runs over all member patents that pass the floor, the page gate and the relevance filter, so that exposure grows with the number of such patents. The alternatives recorded in the experiment log are a patent-level sum over each patent's maximal similarity to the entity's sentences and aggregations of each sentence's best patent link by mean, harmonic mean or maximum, compared on an earlier build. Under the relevance filter an entity may resemble every patent of a technology and still receive zero exposure because it never enters any patent's top ten. Zeros are statements about rank and should be read as such in applications.

C.5 The concrete-task bias

Task sentences written in operational language lie closer to patent text in embedding space irrespective of topic, so that similarity partly reflects linguistic concreteness as well as technological relevance (Section 4.3). Corrections at the level of the measure were evaluated and none removed the bias. The experiment log records the diagnosis.

C.6 Citation weights of recent cohorts

Equation (4) weights each patent by its forward citations relative to the mean of its filing-year cohort, so that an uncited patent carries no weight and the weights of a cohort average one. Table 19 reports the share of uncited patents by filing year among the digital patents tagged to a technology, at the date of the extract. It stays below a tenth for the cohorts filed before 2017 and rises to about half for 2021 and four fifths for 2023, because recent patents have had little time to be cited. The cohort

normalisation keeps the total weight of a year's patents unchanged, so the exposure of a window is not reduced, but within the most recent cohorts it is concentrated on a shrinking minority of patents. The contents of the 2020–2023 period of the employment test rest on that minority.

Table 19: Uncited patents by filing year among the digital patents tagged to a technology.

Filing year	Patents	Uncited (percent)
2008	14,210	6.4
2009	13,560	6.7
2010	14,290	7.3
2011	16,140	7.5
2012	17,619	7.7
2013	18,881	8.3
2014	20,905	7.4
2015	24,312	7.4
2016	31,706	8.3
2017	42,150	9.8
2018	54,552	12.4
2019	71,793	17.7
2020	84,458	27.8
2021	92,791	48.4
2022	103,757	66.6
2023	111,029	78.6

C.7 Record of the corrections

The corrections of the LLM as judge in the digital domain combine two records. The record of the language-model review of the taxonomy is in Appendix B.7. The link review ran in two rounds. The first judged, for every technology and separately for occupations and industries, the five largest sentence-patent pairs, and the second up to the fifty largest links, grouped by patent, carrying over 1,585 links judged in the first round. The cell list was assembled in successive readings of the ranked entity lists of all technologies. It covers word collisions such as refuse sorters on Sorting Algorithms, coding clerks on Demodulation and Detection, garment pattern-makers on Pattern Matching and String Algorithms, the forestry and logging industry on System Logging, storage and warehousing on storage technologies, letter post on Email Systems and translation services on Soft-Decision Decoding. It also covers captures from other domains such as medical occupations on Software Calculators and textile industries on Deformable Body Simulation, and 32 cells of the Quantum Computing technologies, 23 on the occupation side and 9 on the industry side, in which the review judged most links unrelated to the technology or which are plain word collisions. Every green node has exposed entities.

C.8 Further evidence on the structure of the exposures

Figures 23 and 24 report the median exposure over 2009–2023 by technology root or group across industry sections, Figure 25 the correlations between the ten digital roots and the ten green groups

in their exposure across four-digit occupations, and Figure 26 the distribution of each exposure across four-digit occupations. Section 4.4 discusses them.

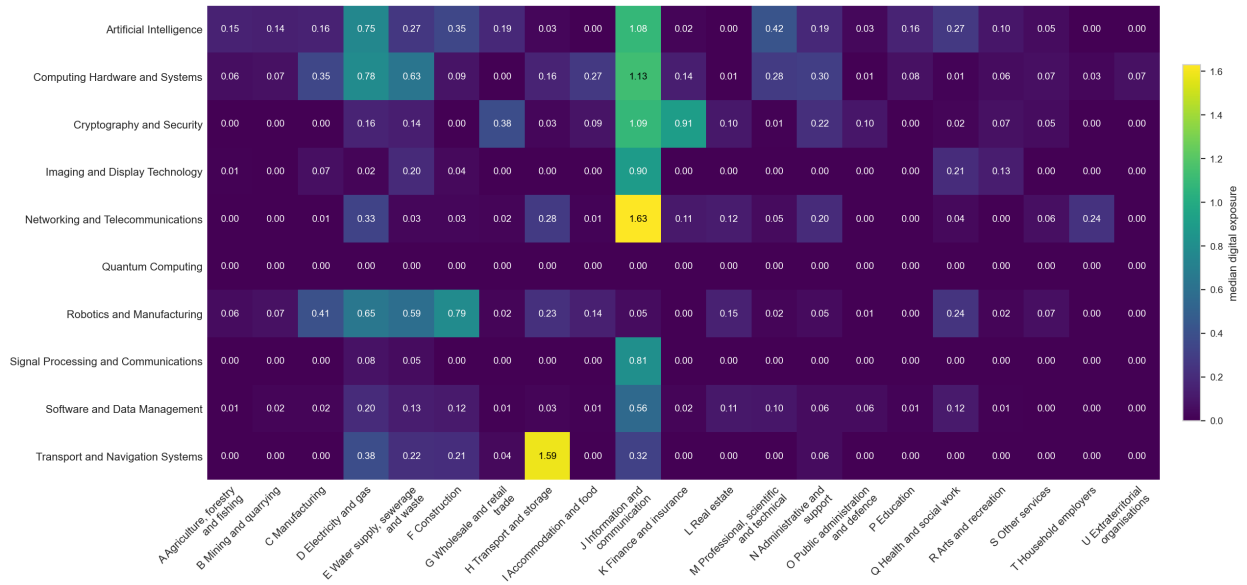


Figure 23: Median digital exposure over 2009–2023 by technology root and NACE Rev 2 section. Information and communication leads for seven of the ten roots. Robotics and manufacturing reaches construction, electricity and gas, water and waste and manufacturing, and transport and navigation reaches transport and storage.

D Design and further results of the direction measure

D.1 The rating prompt

Each call sends a system message, which sets the model's standing instructions, and a user message, which poses the question. The placeholder `{domain}` is digital or green, `{Domain}` its capitalised form at the start of a line, and the bracketed fields are filled from the triplet. The system message is reproduced below.

```
You are an expert in {domain} technologies. Your expertise is defined by the
given persona and context in user prompt.
Task: Given a prompt, produce exactly ONE JSON object evaluating the statement.

STRICT OUTPUT FORMAT:
- Output MUST be a single JSON object with exactly these keys in this order:
  1) "rating": an integer from 0 to 5 (no decimals).
  2) "reason": a 1-3 lines string (<= 25 words), plain text.
- No additional keys, no trailing commas.
- No prose before or after the JSON. No code fences. No comments. No markdown.
```

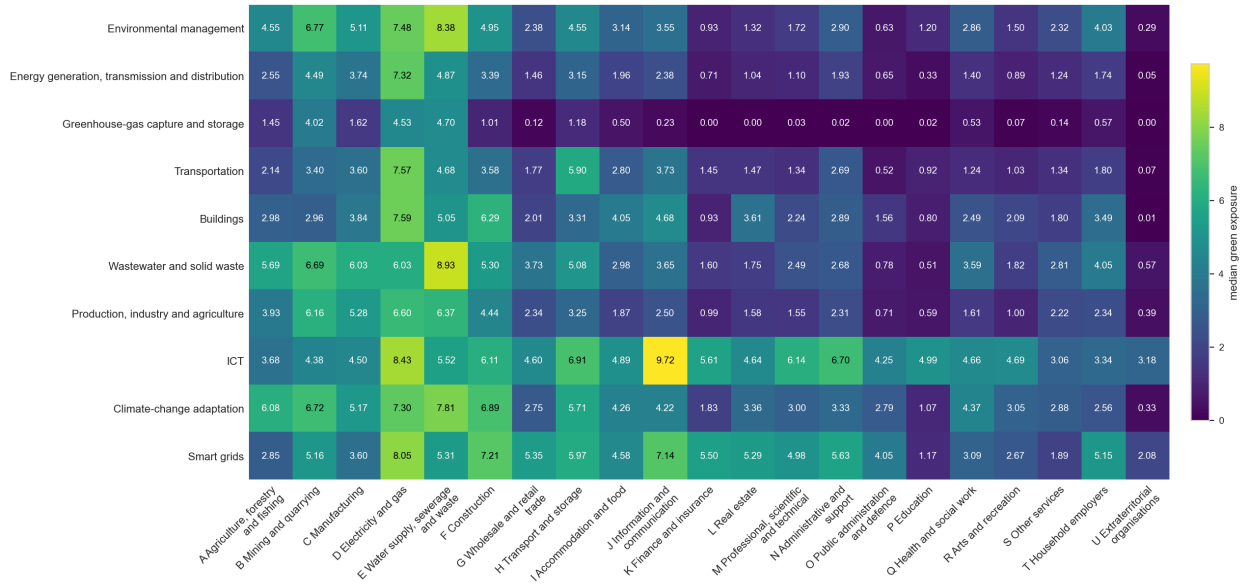


Figure 24: Median green exposure over 2009–2023 by ENV-TECH group and NACE Rev 2 section. Exposure concentrates in electricity and gas, water and waste, mining, construction, manufacturing and transport.

```
POLICY:
- If information is ambiguous, choose the closest rating and explain briefly.
- Never ask questions or request more info.
- Use a neutral, expert tone; avoid hedging unless required by the persona.
- 0 means you do not know. Never use 0 to signal a refusal.
```

The user message is reproduced below.

```
You are assessing how a {domain} technology will affect an occupation as it is
practised within a specific industry.
- Substitute: the {domain} technology takes over the work itself, so fewer
  human workers are needed to do it.
- Complement: the {domain} technology supports or enhances the human workers,
  who still do the work, but better, faster or at higher quality.

You are {role}. Answer strictly from that perspective.

Occupation: [occupation group title]
Industry: [NACE section title]
{Domain} technology: [technology name or ENV-TECH description]

To what extent do you agree with the statement below?
- 1 = Strongly disagree
- 2 = Disagree (25 percent)
- 3 = Neutral / somewhat agree (50 percent)
```

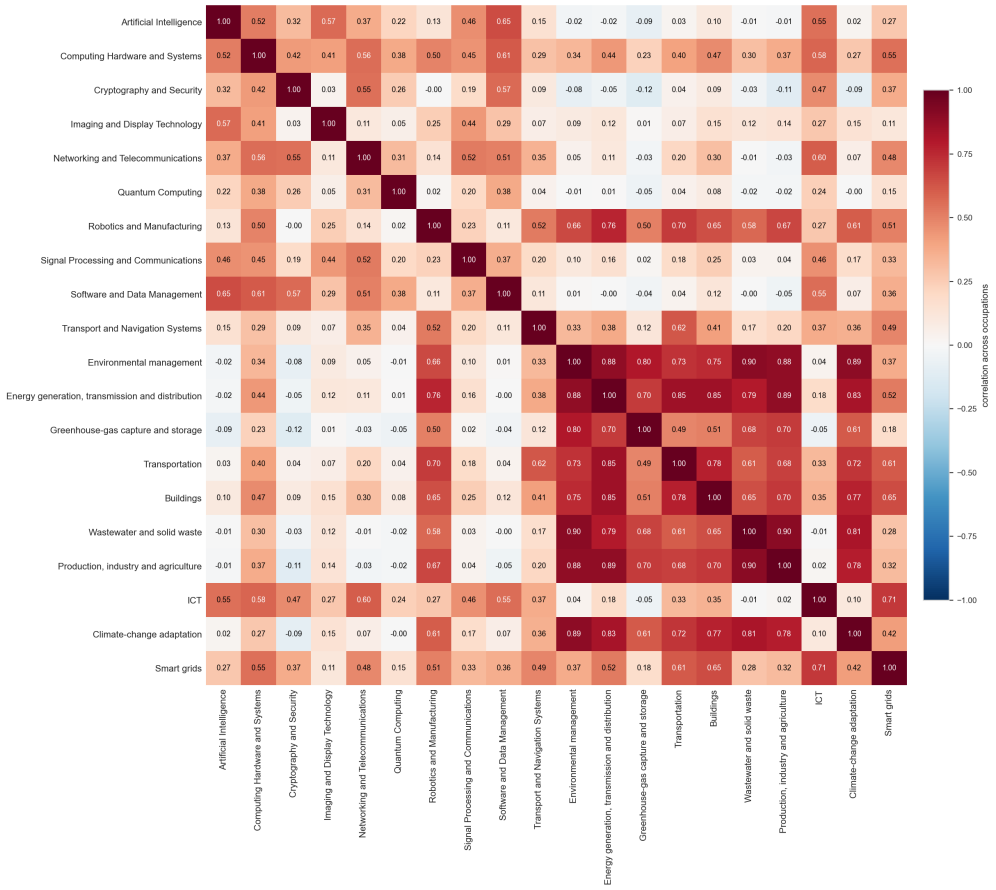


Figure 25: Correlations between the ten digital roots and the ten green groups in exposure across four-digit occupations over 2009–2023. The green groups other than ICT and smart grids form one correlated block. Robotics and manufacturing, computing hardware and transport and navigation correlate with it, and the other digital roots hardly do.

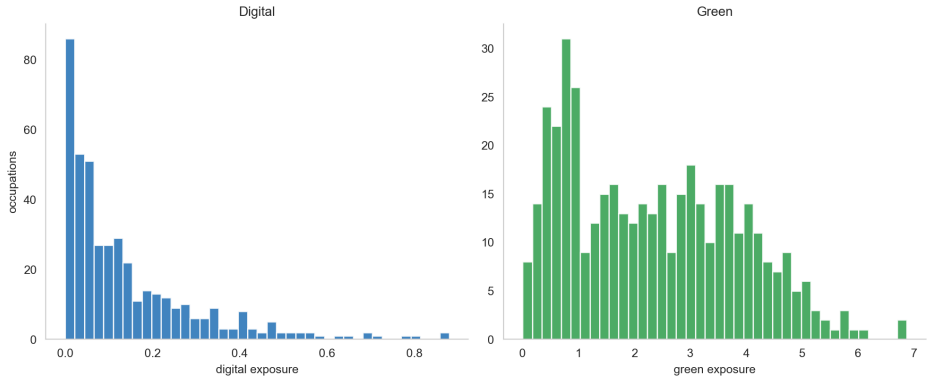


Figure 26: Distribution of digital and green exposure over 2009–2023 across four-digit occupations, in the units of each domain. Digital exposure is concentrated at low values with a long right tail, while green exposure is spread more evenly.

```
- 4 = Mostly agree (75 percent)
- 5 = Fully agree
- 0 = Do not know

Statement: {statement}

Provide exactly one rating from [0, 1, 2, 3, 4, 5] and one concise reason
(1-3 sentences).
```

The role is "a manager who runs the team that performs this work day to day", "an experienced worker who personally performs this task day to day" or "a chief executive officer responsible for the organisation's strategy and investment decisions". The substitution statement is "This {domain} technology will replace human workers in this occupation within this industry." and the complementarity statement "This {domain} technology will work alongside human workers in this occupation within this industry and support them." The response is constrained by a JSON schema with an integer rating from 0 to 5 and a string reason.

D.2 The prompt-format experiment

Whether consolidated prompt formats lose information was tested on twelve triplets drawn from the exposure grid, one central and one peripheral triplet for each of the three highest-exposure technologies of each domain. Each of four formats with identical wording was run with five draws, for 720 calls and 1,440 ratings in total. The four formats are the separated format, with one construct and one persona per call, a format merging the two constructs, a format merging the three personas, and a fully merged format. The experiment was run before the production prompt was fixed, at temperature and top-p of 1.0, with a horizon of 2030, a human-resources manager in place of the chief executive and the zero defined as do not know or no effect, on triplets from an earlier exposure build. The comparison holds the prompt fixed across formats. Table 20 reports it. Agreement with the separated reference declines monotonically with consolidation, and individual cell means shift by up to three scale points. Use of the no-effect category falls when personas share a call, from 7.8 to 0.3 percent overall with merged personas, and on the single no-effect triplet in the test set from 36.7 to 3.3 percent. The latter comparison rests on one triplet and shows the sign of the change without estimating its size. Draw-to-draw variability increases monotonically with consolidation, and the merged-construct format widens the substitution-complementarity gap by half a scale point, which would bias the measured direction directly. The separated format was therefore retained despite its sixfold cost.

D.3 The grid design space

The grid is one point in a design space with six dimensions. These are occupation resolution (three- or four-digit ISCO), industry resolution (NACE section, division or group), technology resolution (base nodes, intermediate levels, or one indicator per domain), the exposure filter (none, top-

Table 20: Prompt-format experiment on twelve triplets, with five draws per format. The separated format is the reference.

	Separated	Merged constructs	Merged personas	Fully merged
Calls per triplet-draw	6	3	2	1
Correlation with separated cell means	1	0.883	0.805	0.764
Mean absolute deviation		0.42	0.52	0.53
Zero share, all ratings (%)	7.8	4.2	0.3	3.1
Zero share, no-effect triplet (%)	36.7	33.3	3.3	10.0
Within-prompt draw SD	0.475	0.523	0.587	0.646
Net rating (substitution minus complementarity), mean	−1.056	−1.572	−0.811	−1.146

K per margin, or percentile cuts), the source recording occupation-industry pairs (the LFS, the OECD Survey of Adult Skills, online vacancies, or none), and the co-occurrence floor. The adopted design combines the finest resolution at which a representative EU-wide employment source records pairs, three-digit occupations by sections under the LFS, with the common exposure thresholds of Section 5.2, which are 1.31 for digital occupations, 1.78 for digital industries, 3.79 for green occupations and 4.15 for green industries. The 0.1 percent employment floor is the co-occurrence cut in the LFS, where presence alone admits pairs with negligible employment. Four-digit occupations could be recorded only through the Survey of Adult Skills (PIAAC), which covers thirteen EU countries and in which any share-based floor reduces to requiring that the pair be observed at all, because the median occupation counts about 140 respondents. The experiment log reports the sizes of the alternative grids.

D.4 Personas

The personas move the level of the ratings and leave their ranking. The chief executive returns the highest mean substitution rating in both domains, 2.38 in the digital and 2.06 in the green, and the worker the lowest, 2.10 and 1.85, while complementarity differs little across roles, from 3.50 to 3.43 in the digital and from 3.07 to 2.88 in the green domain (Figure 27). Across triplets the three persona means correlate between 0.88 and 0.95 and differ in absolute value by 0.17 to 0.35 scale points. In a decomposition of the variance of the individual ratings, between 70 and 84 percent lies between triplets, 4 to 7 percent between the personas of one triplet, and 12 to 23 percent between draws of one persona.

D.5 How the zero category is used

Zeros are 9.4 percent of all ratings, 7.4 and 3.4 percent for digital complementarity and substitution and 13.4 and 9.6 percent for their green counterparts, and the worker uses the category almost twice as often as the chief executive, in 12.1 against 6.9 percent of answers. Although the scale defines the zero as do not know, the justifications, which we read for every zero rating, show that it functions as a no-effect code. Outside the twelve green nodes discussed below, which account for

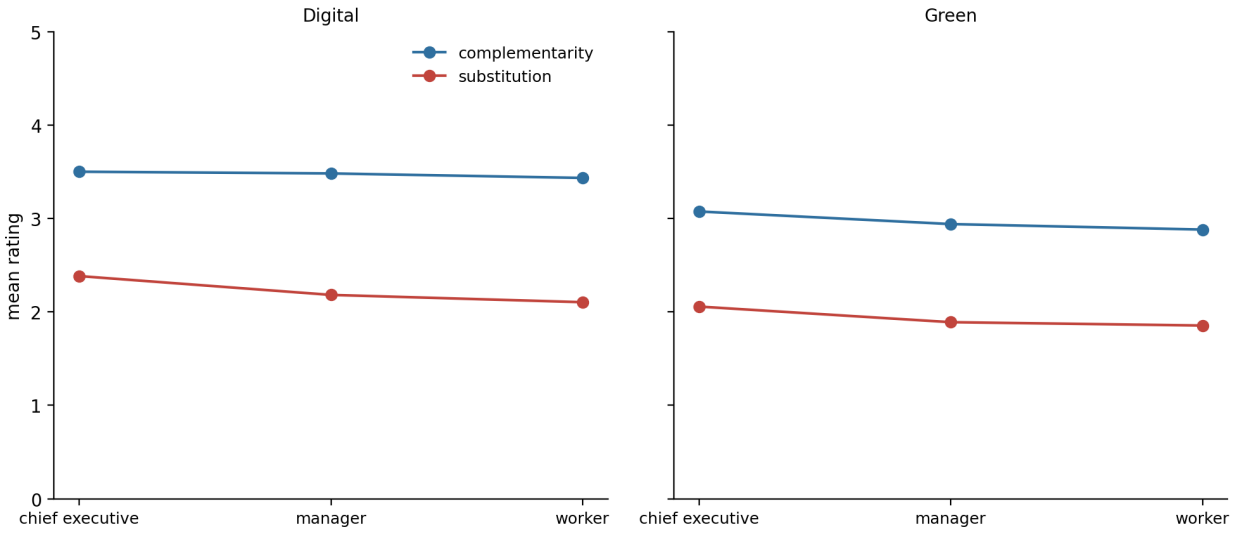


Figure 27: Mean rating by persona, construct and domain. The three roles differ by at most 0.28 scale points on substitution and 0.19 on complementarity, with the chief executive highest on substitution and the worker lowest in both domains.

30.8 percent of the zeros, the stated reason is typically that the technology has no relation to the work of that occupation in that industry, which is the content of a rating of 1. The two categories are interchangeable at the level of the individual response. Appendix D.9 reports the employment test with the second grade, whose label names a quarter, coded as a quarter instead of none. The same persona rating the same triplet returns the same justification as a zero in one draw and as a 1 in another. For additive manufacturing with financial and mathematical associate professionals in accommodation and food service, the worker's zero reads "I see no direct connection between 3D printing and my daily tasks like budgeting or data analysis" and the worker's 1 "I see no direct link between 3D printing and my core tasks like budgeting or revenue analysis. It does not support or enhance my daily accounting or reporting work." A zero is therefore retained as a zero, and S and C are means over all fifteen ratings per construct on a scale running from 0 to 5, with the zero as the bottom category.

The exception is a set of twelve green nodes whose ENV-TECH descriptions are not readable as technologies once detached from their parent heading. Six name an activity that is not itself green, among them conventional vehicles, aeronautics or air transport, glass production and nuclear fission reactors, and six are umbrella descriptions such as enabling technologies with a potential contribution to emissions mitigation or technologies having an indirect contribution to adaptation. For these nodes the zero records that the label cannot be rated and does not mean that the technology leaves the work unaffected. Their zeros are 1.7 and 1.2 percent of all ratings respectively, 2.9 percent together, while the twelve nodes carry 19 percent of all ratings. Appendix D.6 lists them with their zero shares and the justifications on which the reading rests.

D.6 Green nodes not readable as technologies

The justifications of the zero ratings separate the twelve nodes into two groups. In the first, the ENV-TECH description names an activity that is not green once separated from its parent heading on climate change mitigation, and the respondent rejects the premise of the question. For glass production rated for refuse workers in water supply and waste management, one justification reads "Glass production is not a green technology used in waste management; it is a separate manufacturing process." In the second, the description is an umbrella formulation that names no technology. For technologies with an indirect contribution to adaptation rated for drivers in manufacturing, one justification reads "The technology is too vaguely defined to assess its direct impact on driving tasks in manufacturing." Table 21 reports the twelve nodes. The nodes are retained with the ratings as given, and the reading of the zero adopted in Section 5.3 applies to the zeros outside these nodes, about 69 percent of all zeros.

Table 21: The twelve green nodes whose descriptions are not readable as technologies, with their use of the zero category. Together they carry 30.8 percent of all zero ratings, and their zeros are 2.9 percent of all ratings.

Node description	Triplets	Ratings	Zeros	Share (%)
Activity not readable as green without its parent heading				
Conventional vehicles (based on internal combustion engine)	224	6,720	1,775	26.4
Aeronautics or air transport	107	3,210	1,491	46.4
Glass production	43	1,290	883	68.4
Production of cement	22	660	212	32.1
Nuclear fission reactors	85	2,550	1,448	56.8
Ceramic production	23	690	363	52.6
Umbrella description naming no technology				
Technologies having an indirect contribution to adaptation to climate change	379	11,370	1,723	15.2
Enabling technologies with a potential contribution to GHG emissions mitigation	812	24,360	730	3.0
At coastal zones; at river basins	60	1,800	930	51.7
Enabling technologies or technologies with a potential or indirect contribution to GHG emissions mitigation	132	3,960	193	4.9
Technologies in the production process for final industrial or consumer products	322	9,660	587	6.1
Climate change mitigation technologies for sector-wide applications	88	2,640	91	3.4

D.7 The extremes of the distribution

In the digital domain the highest substitution is rated for language models with keyboard operators in administrative and support services, 4.93, and in information and communication, 4.73, followed by robot manipulators with assemblers in manufacturing at 4.67 and by database systems and electronic payment processing with keyboard operators in information and communication and in finance, 4.60 each. The lowest ratings, zero on both constructs, pair semiconductor fabrication with

garment trades workers and with cleaners in other service activities. In the green domain the highest substitution is rated for production process technologies with keyboard operators in manufacturing at 4.27 and for metal processing technologies with mining and mineral processing plant operators in mining and quarrying, one of five triplets at 4.00, while some seventy green triplets are rated zero, such as hydro energy with cleaning workers in transportation. The highest green complementarity, conventional vehicles with car, van and motorcycle drivers at 4.93, illustrates the labelling problem of Appendix D.5 and does not show a green technology supporting drivers.

D.8 Functional form

The employment test of Section 6.3 enters the measure as two contents, Σ_e and Γ_e. This appendix compares it with single indices that combine exposure and direction into one number per occupation. Each index combines X_e, the percentile rank of the occupation's domain exposure, with the substitution share $\pi_e = \Sigma_e/(\Sigma_e + \Gamma_e)$ and the complementarity share $\theta_e = 1 - \pi_e$. The shares are defined for the 372 digital and 417 green four-digit occupations with rated content. The indices are

X_e	exposure alone,
$X_e\,\pi_e$	the product,
$X_e\,\big[1 - w\,(\theta_e - \theta_{\min})\big], \quad w \in \{0.25,\ 0.5,\ 1\}$	the IMF adjustment,
$X_e\,\exp\big[-(\theta_e - \theta_{\min})\big]$	the exponential IMF adjustment,
$M_\rho = \big(\tfrac{1}{2}X_e^\rho + \tfrac{1}{2}\pi_e^\rho\big)^{1/\rho}, \quad \rho \in \{1,\ 0.5,\ 0,\ -1,\ -5,\ -\infty\}$	the power mean,

where $\theta_{\min}$ is the smallest complementarity share across occupations. The IMF adjustment follows Pizzinelli, Panton, Mendes Tavares, Cazzaniga, and Li (2023), who scale exposure down by a normalised complementarity index, with $w = 0$ the unadjusted exposure and $w = 1$ the full adjustment. The power mean is the arithmetic mean of X_e and π_e at $\rho = 1$, their geometric mean $\sqrt{X_e\pi_e}$ at $\rho = 0$, their harmonic mean at $\rho = -1$ and their minimum at $\rho = -\infty$, in which the lower of the two binds.

Table 22 reports the rank correlation of each index with the product across occupations. The geometric mean ranks occupations as the product does by construction. The correlation is 0.89 or higher for every other power mean and for the full IMF adjustment in both domains. The partial IMF adjustments lie between exposure alone and the product, at 0.76 to 0.85 for digital and 0.85 to 0.90 for green, and exposure alone correlates at 0.73 and 0.83. Direction therefore changes the ranking of occupations, and among the power means and the full IMF adjustment the ranking hardly depends on how direction is combined with exposure. Exposure and direction are close to independent across occupations in the digital domain and associated in the green domain. Of the 186 digital occupations with above-median exposure, 96 have an above-median substitution share, against 93 under independence, and of the 209 green ones 130, against about 105.

Tables 23 and 24 enter each index as the only exposure regressor in the design of Table 7, averaged

over the four-digit occupations of each group with rated content and built, like the contents, from the patents filed in the years of each outcome period. On the same cells the two contents change by less than 0.01. Exposure alone is not significant in any period or source, and no IMF adjustment is significant for digital. For digital over 2020–2023, the product and the power means with $\rho \leq 0.5$ are negative and significant in both sources. Over 2011–2023 in the microdata the product is -0.091, significant at 10 percent, and no other index is significant, while the two contents are -0.125 and 0.202, both significant at 1 percent. For green, only the product in both sources and the full IMF adjustment in the microdata reach the 10 percent level, over 2020–2023. No index is significant over 2011–2019. A single index cannot separate the two contents, which enter with opposite signs.

Table 22: Rank correlation (Spearman) of each index of exposure X (percentile rank) and substitution share π with the product $X\pi$, across the four-digit occupations with rated content. The geometric mean ranks occupations as the product does by construction.

Form	Digital	Green
Exposure alone, X	0.73	0.83
IMF adjustment, $w = 0.25$	0.76	0.85
IMF adjustment, $w = 0.5$	0.81	0.87
IMF adjustment, $w = 1$	0.89	0.92
IMF adjustment, exponential	0.85	0.90
Power mean, $\rho = 1$ (arithmetic)	0.92	0.95
Power mean, $\rho = 0.5$	0.98	0.98
Power mean, $\rho \to 0$ (geometric)	1.00	1.00
Power mean, $\rho = -1$ (harmonic)	0.97	0.98
Power mean, $\rho = -5$	0.92	0.91
Power mean, $\rho \to -\infty$ (minimum)	0.91	0.90
Occupations	372	417

D.9 Exposure, direction and the coding of the scale

Table 25 re-estimates the employment test of Table 7 in three variants. The first enters the domain exposure of the occupation group alone. The second enters it next to the two contents. The third keeps the two contents and codes the second grade of the scale, whose label names a quarter, as a quarter of the reached content instead of none. Exposure alone is not associated with employment change in any period or source. Entered next to the contents, with which it correlates at 0.7 to 0.9, it leaves substitution significant over 2020–2023 in both sources and complementarity in the microdata, while over 2011–2023 in the microdata the three regressors separate less cleanly, with exposure negative and substitution no longer significant. The alternative coding of the second grade makes the contents larger, strengthens the digital and the green associations over 2011–2023 and 2020–2023, and leaves 2011–2019 without a significant substitution content.

Table 23: Single indices of digital exposure and direction and employment change. Each index of Table 22 is the only exposure regressor in the design of Table 7.

	Microdata, ISCO three-digit			Public table, ISCO two-digit		
Change in log employment	2011–2019 (1)	2020–2023 (2)	2011–2023 (3)	2011–2019 (4)	2020–2023 (5)	2011–2023 (6)
Substitution content (Table 7)	0.009	-0.081***	-0.125***	0.055	-0.044**	-0.023
	(0.027)	(0.017)	(0.046)	(0.051)	(0.021)	(0.059)
Complementarity content (Table 7)	0.055*	0.067***	0.196***	0.010	0.033**	0.064
	(0.033)	(0.013)	(0.048)	(0.036)	(0.015)	(0.043)
Substitution content, same cells	0.009	-0.082***	-0.125***	0.056	-0.045**	-0.023
	(0.027)	(0.017)	(0.047)	(0.053)	(0.021)	(0.059)
Complementarity content, same cells	0.056	0.066***	0.202***	0.005	0.034**	0.062
	(0.038)	(0.015)	(0.054)	(0.037)	(0.014)	(0.044)
Exposure alone, X	0.027	0.000	0.030	0.026	-0.000	0.026
	(0.034)	(0.024)	(0.055)	(0.034)	(0.021)	(0.042)
Product, $X\pi$	-0.017	-0.057***	-0.091*	0.014	-0.041**	-0.031
	(0.030)	(0.018)	(0.050)	(0.034)	(0.020)	(0.038)
IMF adjustment, $w = 0.25$	0.024	-0.005	0.019	0.026	-0.003	0.022
	(0.034)	(0.024)	(0.056)	(0.034)	(0.022)	(0.042)
IMF adjustment, $w = 0.5$	0.020	-0.012	0.005	0.026	-0.006	0.018
	(0.035)	(0.024)	(0.056)	(0.035)	(0.022)	(0.042)
IMF adjustment, $w = 1$	0.009	-0.029	-0.027	0.024	-0.016	0.006
	(0.034)	(0.022)	(0.055)	(0.036)	(0.023)	(0.042)
IMF adjustment, exponential	0.014	-0.021	-0.013	0.026	-0.010	0.013
	(0.035)	(0.023)	(0.056)	(0.035)	(0.023)	(0.043)
Power mean, $\rho = 1$ (arithmetic)	0.006	-0.027	-0.034	0.003	-0.023	-0.018
	(0.031)	(0.021)	(0.053)	(0.029)	(0.022)	(0.035)
Power mean, $\rho = 0.5$	0.007	-0.035*	-0.032	0.007	-0.034*	-0.026
	(0.027)	(0.019)	(0.049)	(0.028)	(0.020)	(0.030)
Power mean, $\rho \to 0$ (geometric)	0.007	-0.041**	-0.030	0.011	-0.044**	-0.034
	(0.024)	(0.019)	(0.047)	(0.027)	(0.017)	(0.027)
Power mean, $\rho = -1$ (harmonic)	0.006	-0.048***	-0.045	0.014	-0.051***	-0.041
	(0.021)	(0.018)	(0.045)	(0.027)	(0.017)	(0.027)
Power mean, $\rho = -5$	0.006	-0.050***	-0.048	0.017	-0.054***	-0.044
	(0.019)	(0.018)	(0.043)	(0.028)	(0.017)	(0.029)
Power mean, $\rho \to -\infty$ (minimum)	0.006	-0.050***	-0.046	0.017	-0.054***	-0.043
	(0.018)	(0.018)	(0.043)	(0.028)	(0.017)	(0.029)
Cells, first two rows		1,739			786	
Cells, other rows		1,566			765	

Change in log employment of country-by-occupation-group cells in the design of Table 7. The first two rows repeat that table. The next two re-estimate it on the cells of the occupation groups with rated content, on which the indices are defined. Each index is the only exposure regressor, averaged over the four-digit occupations of the group with rated content. Exposure and contents are built from the patents filed in the years of the same period. Every regressor is divided by its standard deviation across cells. Columns (1) to (3) use the LFS microdata and columns (4) to (6) the public Eurostat table. Standard errors clustered by occupation group are in parentheses. * p<0.10, ** p<0.05, *** p<0.01.

Table 24: Single indices of green exposure and direction and employment change. Each index of Table 22 is the only exposure regressor in the design of Table 7.

	Microdata, ISCO three-digit			Public table, ISCO two-digit		
Change in log employment	2011–2019 (1)	2020–2023 (2)	2011–2023 (3)	2011–2019 (4)	2020–2023 (5)	2011–2023 (6)
Substitution content (Table 7)	0.022	-0.046	-0.088	-0.038	-0.047**	-0.088**
	(0.031)	(0.029)	(0.066)	(0.028)	(0.020)	(0.037)
Complementarity content (Table 7)	0.003	0.007	0.067	0.061*	-0.001	0.068
	(0.039)	(0.036)	(0.068)	(0.034)	(0.029)	(0.045)
Substitution content, same cells	0.022	-0.048*	-0.092	-0.038	-0.047**	-0.088**
	(0.031)	(0.029)	(0.066)	(0.028)	(0.020)	(0.037)
Complementarity content, same cells	0.004	0.011	0.076	0.061*	-0.001	0.068
	(0.039)	(0.036)	(0.069)	(0.034)	(0.029)	(0.045)
Exposure alone, X	0.023	-0.034	-0.020	0.052	-0.029	0.014
	(0.025)	(0.025)	(0.048)	(0.036)	(0.028)	(0.041)
Product, $X\pi$	0.009	-0.044*	-0.071	0.018	-0.040*	-0.020
	(0.024)	(0.023)	(0.055)	(0.020)	(0.023)	(0.032)
IMF adjustment, $w = 0.25$	0.023	-0.037	-0.025	0.052	-0.032	0.012
	(0.026)	(0.025)	(0.050)	(0.037)	(0.029)	(0.042)
IMF adjustment, $w = 0.5$	0.023	-0.040	-0.032	0.052	-0.035	0.010
	(0.026)	(0.026)	(0.052)	(0.037)	(0.030)	(0.043)
IMF adjustment, $w = 1$	0.021	-0.045*	-0.048	0.049	-0.040	0.002
	(0.027)	(0.026)	(0.056)	(0.035)	(0.030)	(0.043)
IMF adjustment, exponential	0.022	-0.043	-0.040	0.051	-0.038	0.006
	(0.026)	(0.026)	(0.054)	(0.037)	(0.030)	(0.043)
Power mean, $\rho = 1$ (arithmetic)	0.016	-0.039	-0.047	0.049	-0.028	0.021
	(0.027)	(0.026)	(0.057)	(0.034)	(0.032)	(0.042)
Power mean, $\rho = 0.5$	0.015	-0.041	-0.054	0.042	-0.025	0.019
	(0.027)	(0.026)	(0.059)	(0.029)	(0.030)	(0.041)
Power mean, $\rho \to 0$ (geometric)	0.013	-0.039	-0.057	0.031	-0.021	0.014
	(0.025)	(0.024)	(0.058)	(0.025)	(0.027)	(0.038)
Power mean, $\rho = -1$ (harmonic)	0.012	-0.033	-0.050	0.018	-0.018	0.007
	(0.023)	(0.022)	(0.054)	(0.022)	(0.023)	(0.034)
Power mean, $\rho = -5$	0.012	-0.022	-0.034	0.012	-0.009	0.010
	(0.021)	(0.021)	(0.049)	(0.023)	(0.021)	(0.033)
Power mean, $\rho \to -\infty$ (minimum)	0.012	-0.017	-0.025	0.014	-0.006	0.016
	(0.021)	(0.021)	(0.049)	(0.025)	(0.020)	(0.033)
Cells, first two rows		1,739			786	
Cells, other rows		1,711			786	

Change in log employment of country-by-occupation-group cells in the design of Table 7. The first two rows repeat that table. The next two re-estimate it on the cells of the occupation groups with rated content, on which the indices are defined. Each index is the only exposure regressor, averaged over the four-digit occupations of the group with rated content. Exposure and contents are built from the patents filed in the years of the same period. Every regressor is divided by its standard deviation across cells. Columns (1) to (3) use the LFS microdata and columns (4) to (6) the public Eurostat table. Standard errors clustered by occupation group are in parentheses. * p<0.10, ** p<0.05, *** p<0.01.

Table 25: Employment change on exposure alone, on exposure with the two contents, and on the contents with the second grade coded as a quarter.

	Microdata, ISCO three-digit			Public table, ISCO two-digit		
Change in log employment	2011–2019	2020–2023	2011–2023	2011–2019	2020–2023	2011–2023
	(1)	(2)	(3)	(4)	(5)	(6)
A. Digital						
Digital exposure alone	0.022	0.006	0.012	0.040	0.005	0.036
	(0.030)	(0.020)	(0.059)	(0.030)	(0.020)	(0.041)
Digital exposure, with the contents	-0.273**	-0.077	-0.536**	-0.072	0.026	-0.038
	(0.137)	(0.063)	(0.220)	(0.142)	(0.052)	(0.168)
Substitution content, with exposure	0.046	-0.067***	-0.032	0.069	-0.051**	-0.014
	(0.034)	(0.021)	(0.066)	(0.066)	(0.022)	(0.075)
Complementarity content, with exposure	0.289**	0.133**	0.658***	0.070	0.012	0.094
	(0.123)	(0.053)	(0.195)	(0.115)	(0.044)	(0.138)
Substitution content, second grade as a quarter	0.023	-0.180***	-0.324***	0.140	-0.089*	-0.040
	(0.068)	(0.037)	(0.114)	(0.128)	(0.049)	(0.146)
Complementarity content, second grade as a quarter	0.038	0.185***	0.424***	-0.087	0.088**	0.087
	(0.072)	(0.033)	(0.111)	(0.117)	(0.041)	(0.132)
B. Green						
Green exposure alone	0.010	-0.037	-0.037	0.046	-0.038	0.003
	(0.023)	(0.025)	(0.047)	(0.034)	(0.028)	(0.041)
Green exposure, with the contents	-0.014	-0.058	-0.137	0.057	-0.047	-0.030
	(0.053)	(0.037)	(0.094)	(0.073)	(0.032)	(0.061)
Substitution content, with exposure	0.019	-0.058**	-0.119*	-0.021	-0.060***	-0.096**
	(0.034)	(0.029)	(0.071)	(0.023)	(0.021)	(0.037)
Complementarity content, with exposure	0.020	0.071	0.218	-0.001	0.049	0.100
	(0.077)	(0.052)	(0.133)	(0.073)	(0.038)	(0.068)
Substitution content, second grade as a quarter	0.011	-0.125**	-0.299**	-0.081	-0.101**	-0.194**
	(0.068)	(0.058)	(0.130)	(0.061)	(0.044)	(0.085)
Complementarity content, second grade as a quarter	0.013	0.086	0.272**	0.108*	0.053	0.177**
	(0.072)	(0.062)	(0.127)	(0.062)	(0.047)	(0.085)
Observations	1,739	1,739	1,739	786	786	786

Change in log employment of country-by-occupation cells in the design of Table 7, with the same cells, fixed effects, control, weights and standard errors. The domain exposure is the occupation group's mean exposure over the domain's technologies, built from the patents of the same period and divided by its standard deviation across cells. The first two rows of each panel enter exposure alone and then next to the two contents, whose coefficients follow. The last two rows re-estimate the contents of Table 7 with the second grade of the scale coded as a quarter of the reached content instead of none. * p<0.10, ** p<0.05, *** p<0.01.

D.10 Comparison with a usage-based index

Among occupation-level indices of the automation-augmentation distinction, Gmyrek et al. (2023) classify ISCO-08 occupations by automation and augmentation potential, and the generative-AI usage measure of Tomlinson et al. (2025), compared here, separates, for each occupation, the work activities in which a general-purpose AI assistant helps the user from those it performs itself. Mapped to four-digit ISCO codes through the crosswalk from the US Standard Occupational Classification (SOC), the share of activities the assistant performs has a rank correlation of -0.23 with the substitution share across 331 occupations, so the two indices are weakly negatively related. The usage index records what one conversational assistant does in its users' conversations, and the substitution share a judgement about all technologies of a domain in a given industry. The comparison records a divergence between different objects and does not validate the measure.